

GOC

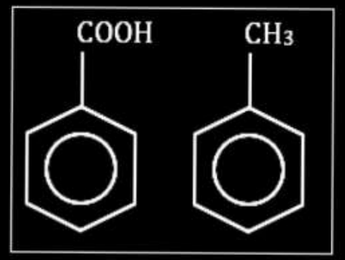
SKM SIR (SURENDRA MISHRA SIR)

Organic Chemistry

Seprate followiy binay mixtre.

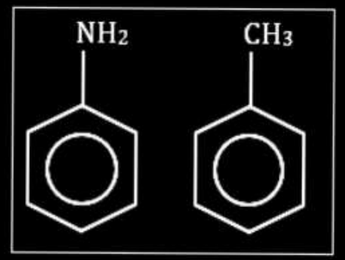
...ture

(19)



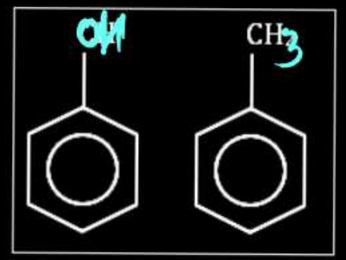
$\xrightarrow[\text{or}]{NaOH}$
$NaHCO_3$

(20)



$\xrightarrow[\text{or}]{Aq.HCl}$
Aq HBr or H_2SO_4
or
Aq HI

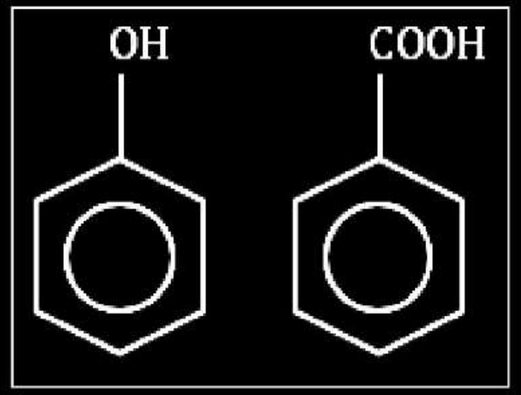
(21)



$\xrightarrow[\text{or}]{Aq.NaOH}$
Aq KOH
Aq. Na_2CO_3

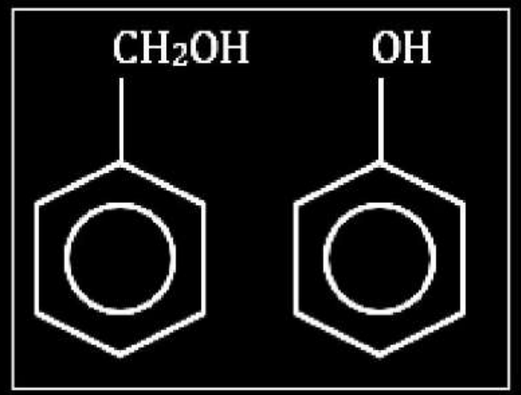
Organic Chemistry

(22)



$Aq.\ NaHCO_3$ →

(23)



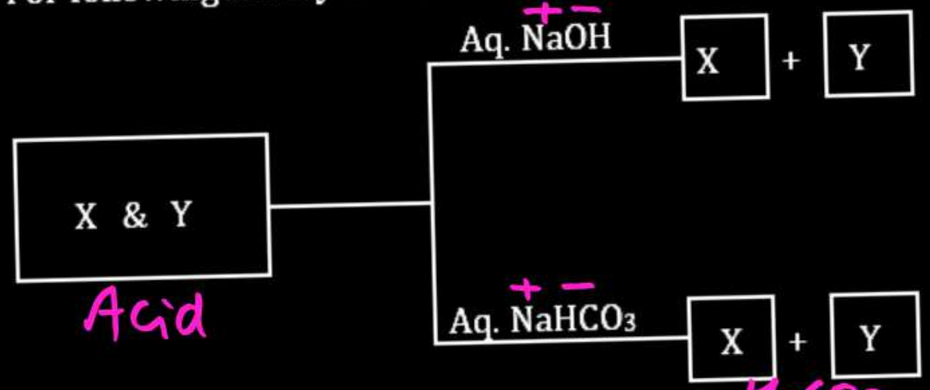
$Aq.\ NaOH$ →

Organic Chemistry

$Y > H_2O > X$

$X > H_2O > Y$

(24) For following Binary mixture



$X > H_2CO_3 > Y$ (or) $Y > H_2CO_3 > X$ (E) N.O.T.

Possible X, Y is

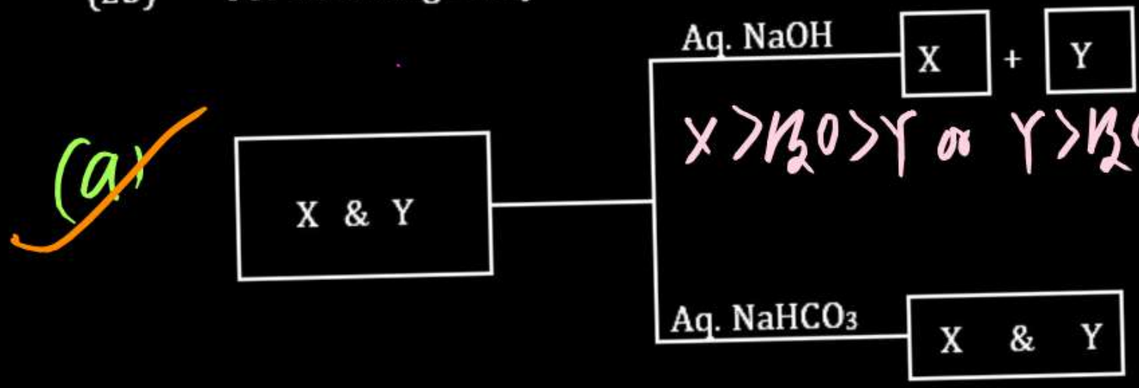
(A) Ph–OH, Ph–CH_2–OH

(B) Ph–OH, Ph–COOH

(C) Ph–CH_2–OH, Ph–COOH

(D) Ph–CH_2–OH, 

(25) For following Binay mix



(a)

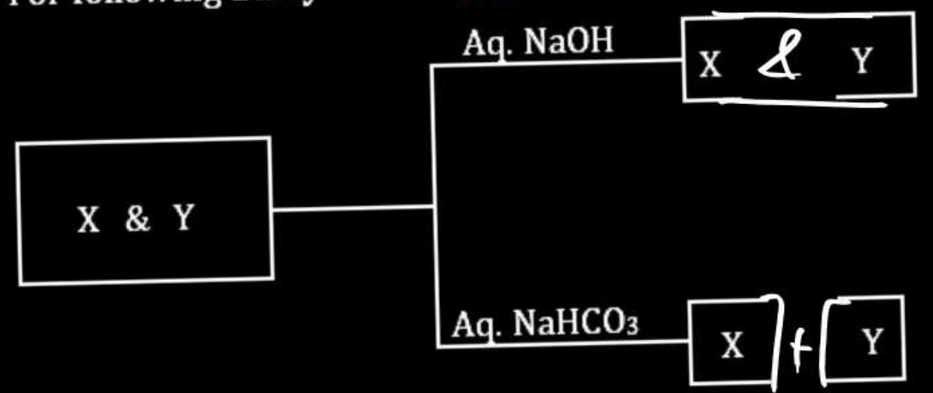
$X > H_2O > Y$ or $Y > H_2O > X$

Ans (A)

$X, Y > H_2CO_3$ or $H_2CO_3 > X, Y$

(25) For following Binay mix

(b)



X, Y > H_2O (or) H_2O > X, Y

Ans (B)

X > H_2CO_3 > Y (or) Y > H_2CO_3 > X

Organic Chemistry

Arrange following in order of acidic strength:

(26) HF < HCl < H Br < HI (unstable)

Conj. Base: $F^{\ominus}$ $Cl^{\ominus}$ $Br^{\ominus}$ $I^{\ominus}$ most stable

IV > III > II > I (stability)

(27) ROH RSH

$RO^{\ominus}$ $RS^{\ominus}$

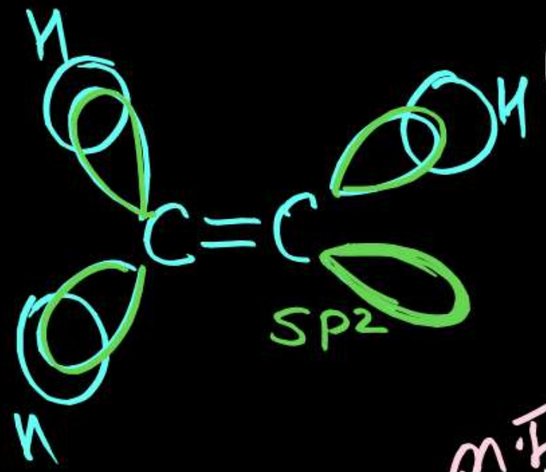
(II > I)

(28) CH_4 NH_3 HF H_2O

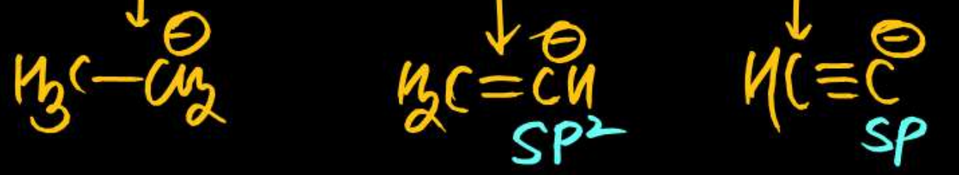
$CH_3^{\ominus}$ $NH_2^{\ominus}$ $F^{\ominus}$ $OH^{\ominus}$

(III > IV > II > I)

Organic Chemistry

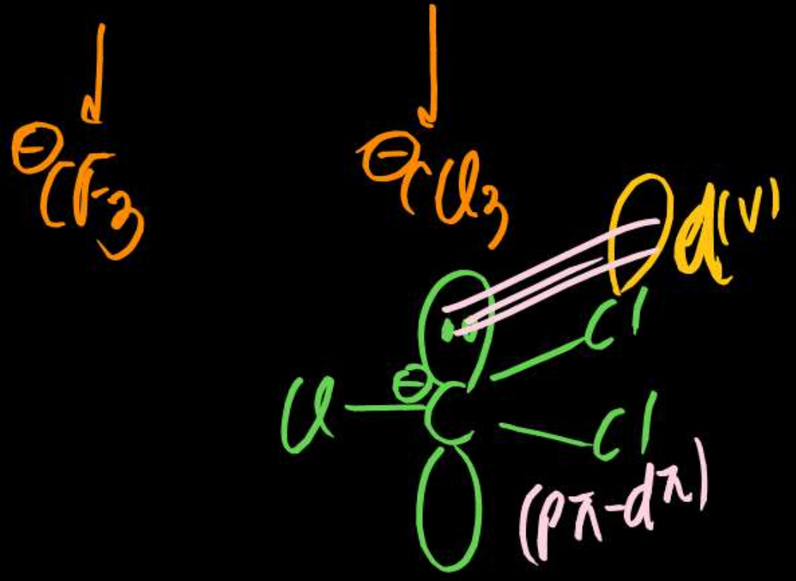


(29) $H_3C - CH_3$ $H_2C = CH_2$ $HC \equiv CH$



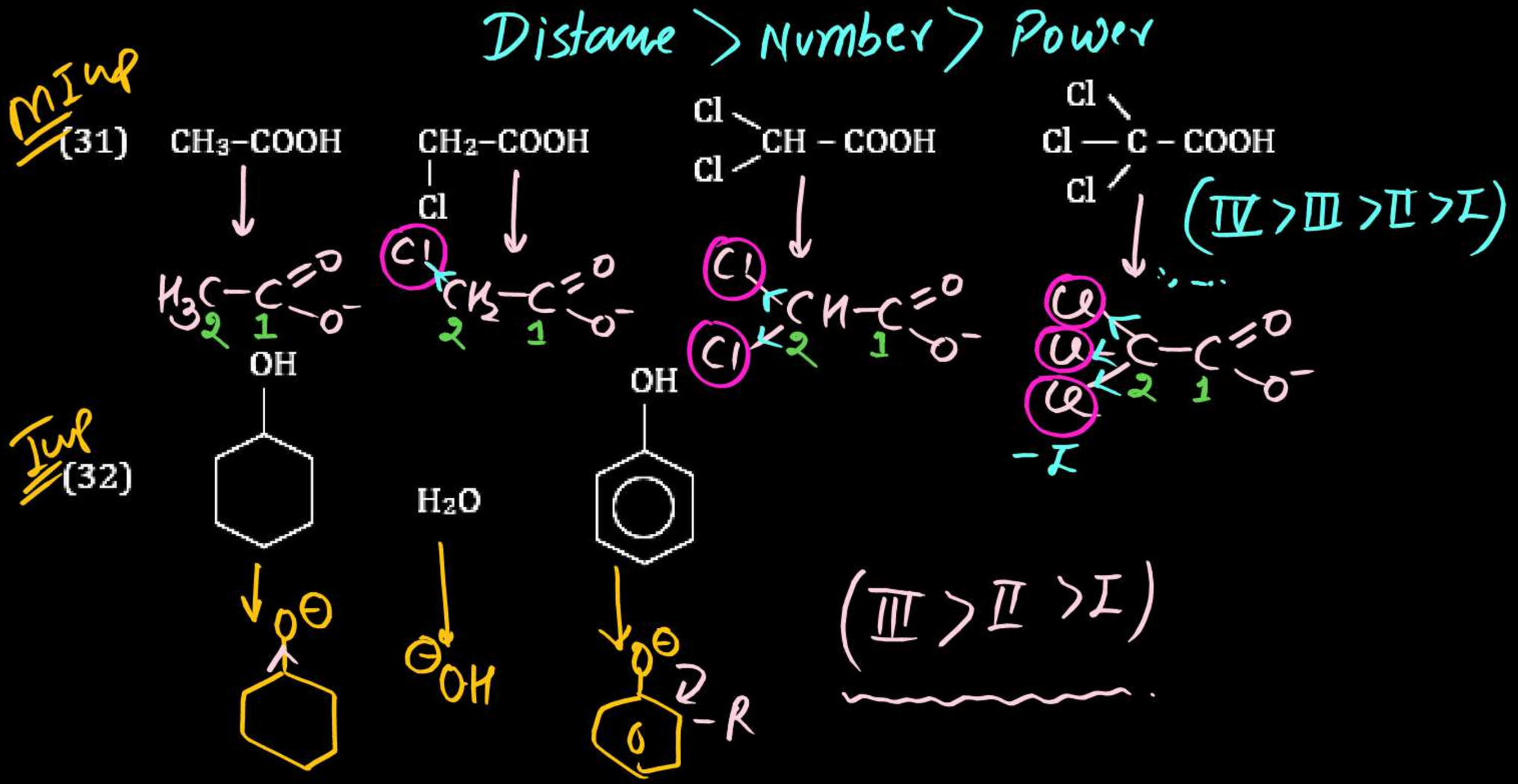
(III > II > I)

m.Imp ~~(30)~~ CHF_3 $CHCl_3$



(II > I)

Organic Chemistry



Organic Chemistry

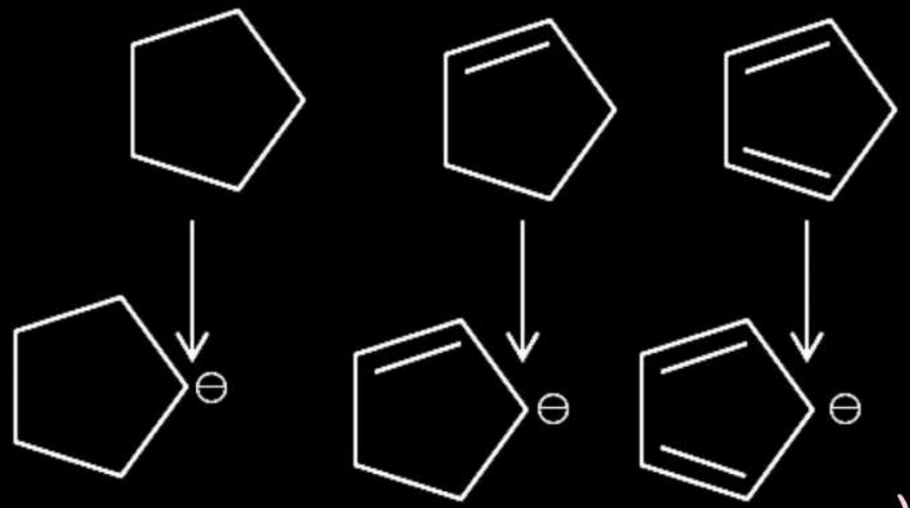
(33) $H_3C-C\equiv CH$ $HC\equiv CH$ $CH_2=CH_2$

C: Base

$H_3C-C\equiv C^{\ominus}$ (sp) $HC\equiv C^{\ominus}$ (sp) $H_2C=CH^{\ominus}$ (sp^2)

II > I > III

(34)

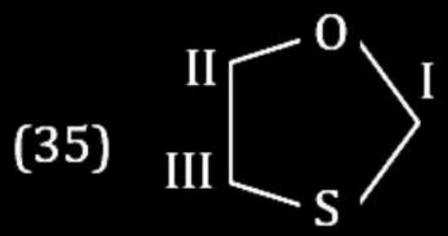


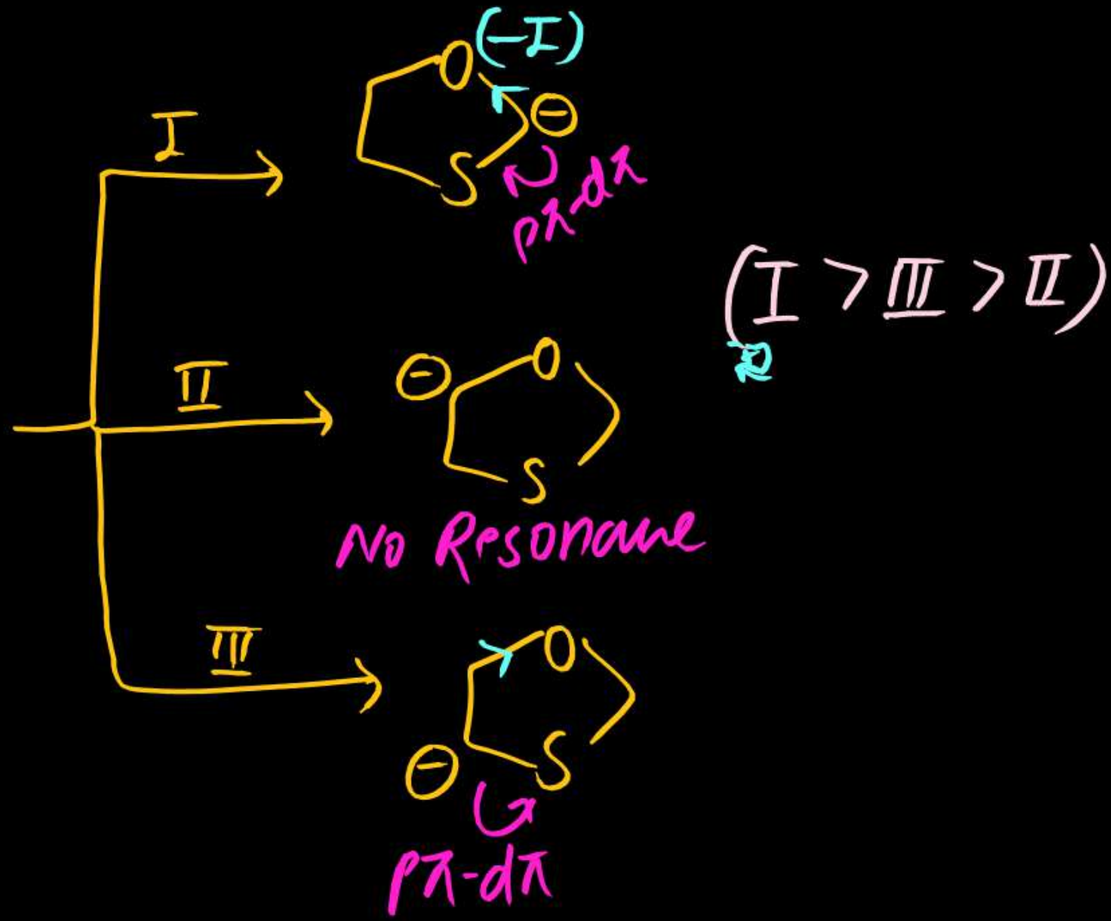
(III > II > I)

(Aromatic)

Organic Chemistry

(35)

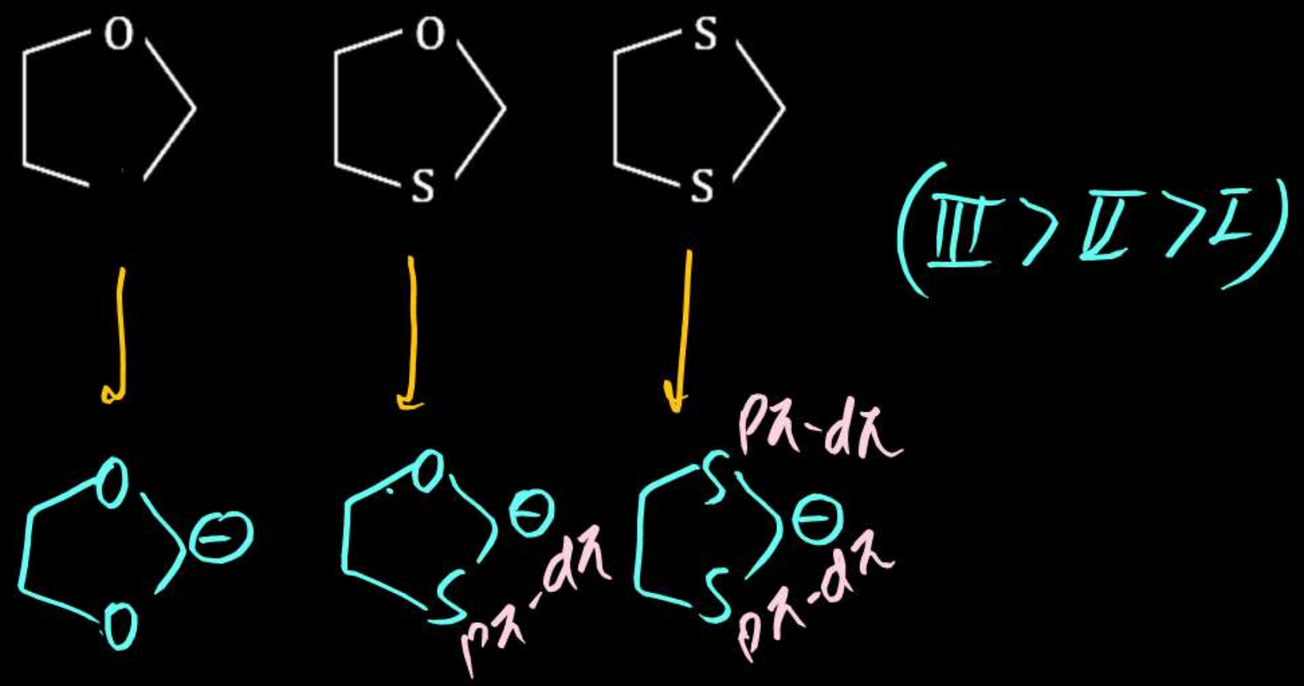




(I > III > II)

Organic Chemistry

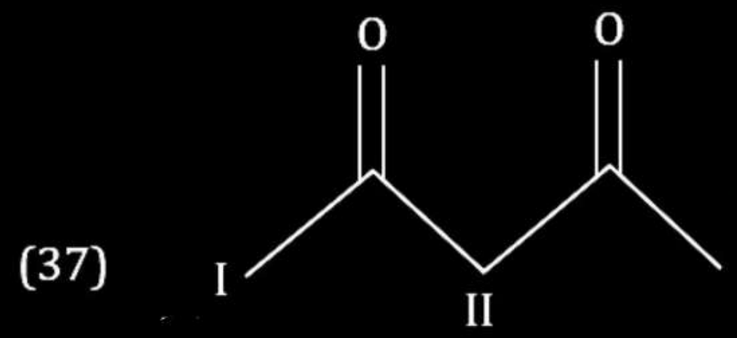
(36)

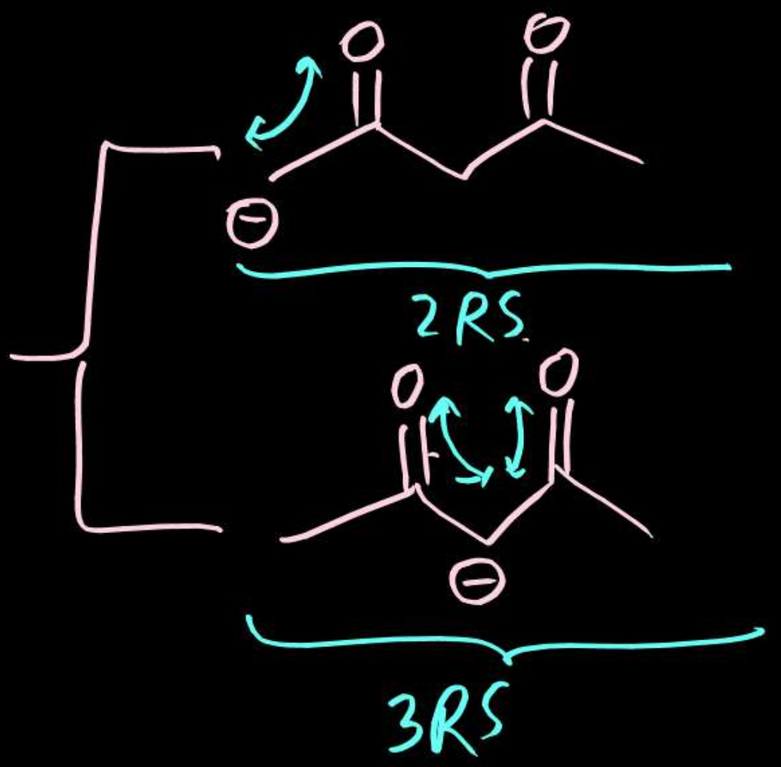


(III > II > I)

Organic Chemistry

(37)

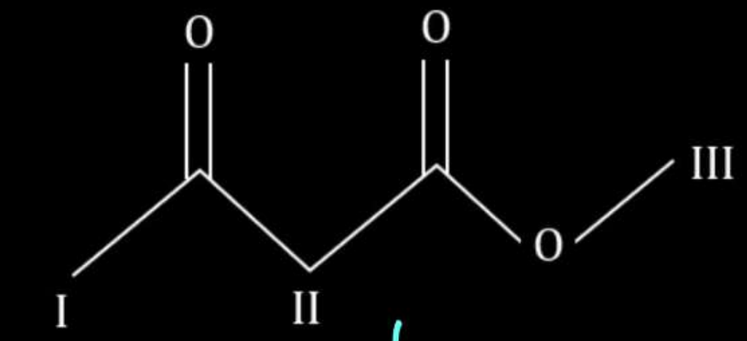




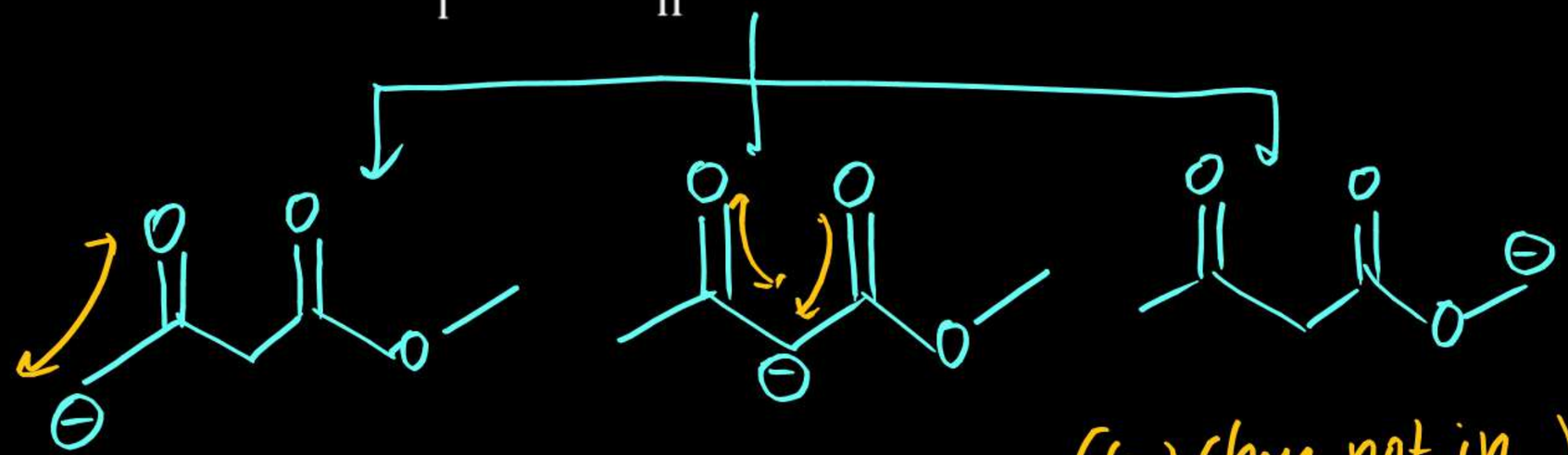
(II > I)

Organic Chemistry

(38)



(II > I > III)



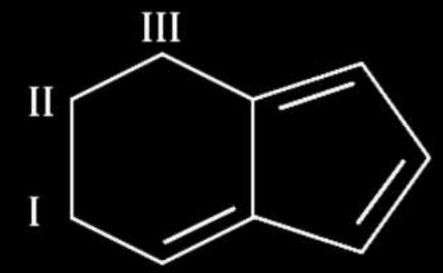
(2RS)

(3RS)

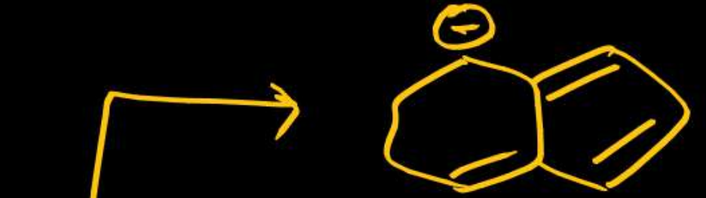
((−) charge not in Resonance)

Organic Chemistry

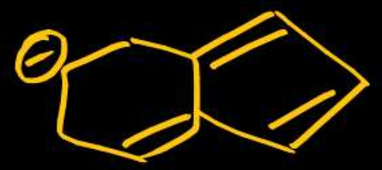
(39)



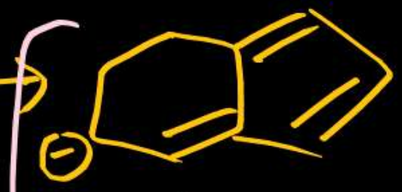
(I > III > II)



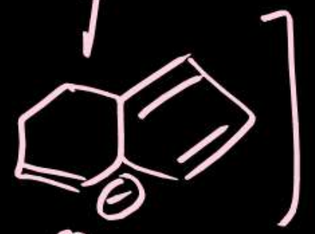
(4RS)



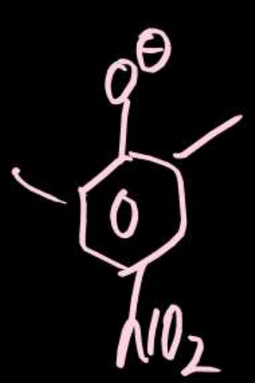
(No Resonance of (-)ve charge)

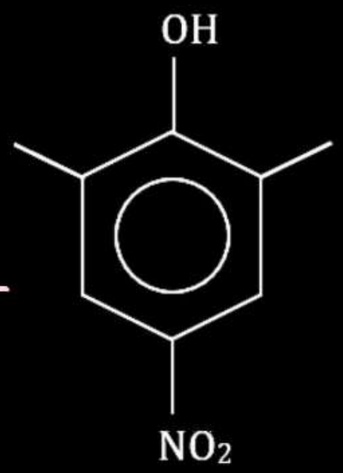


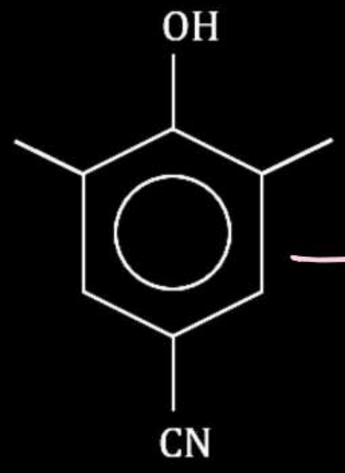
(6RS)

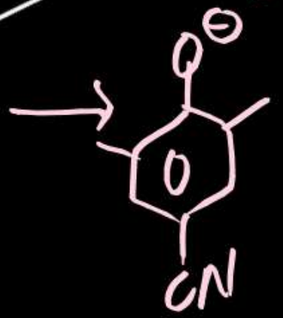


(40)



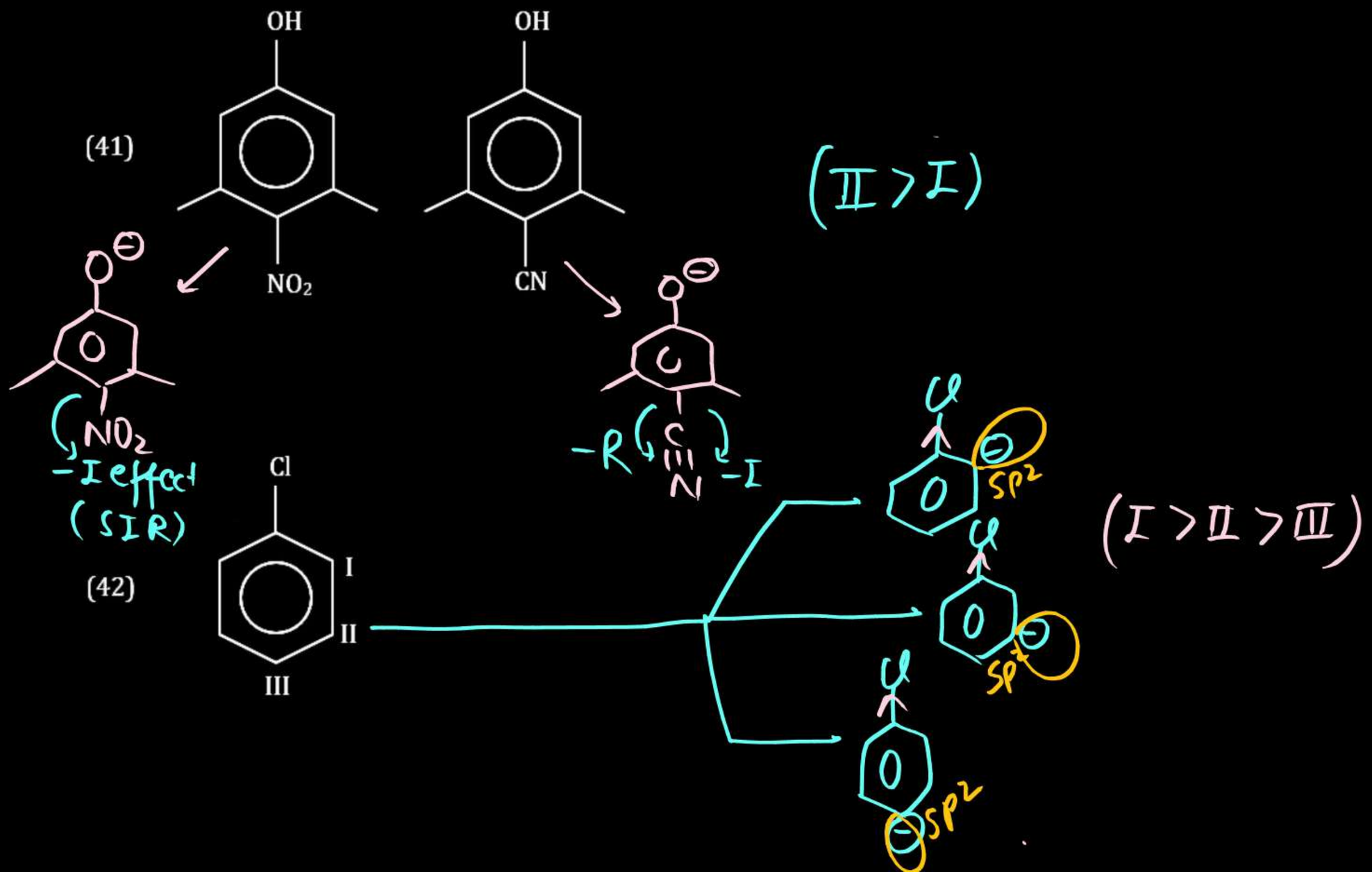






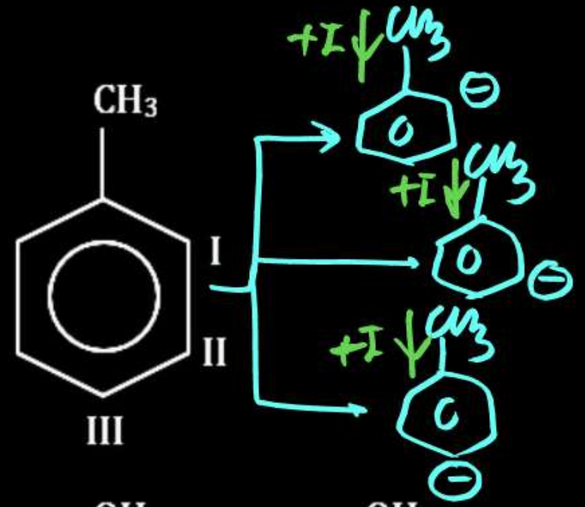
(I > II)

Organic Chemistry



Organic Chemistry

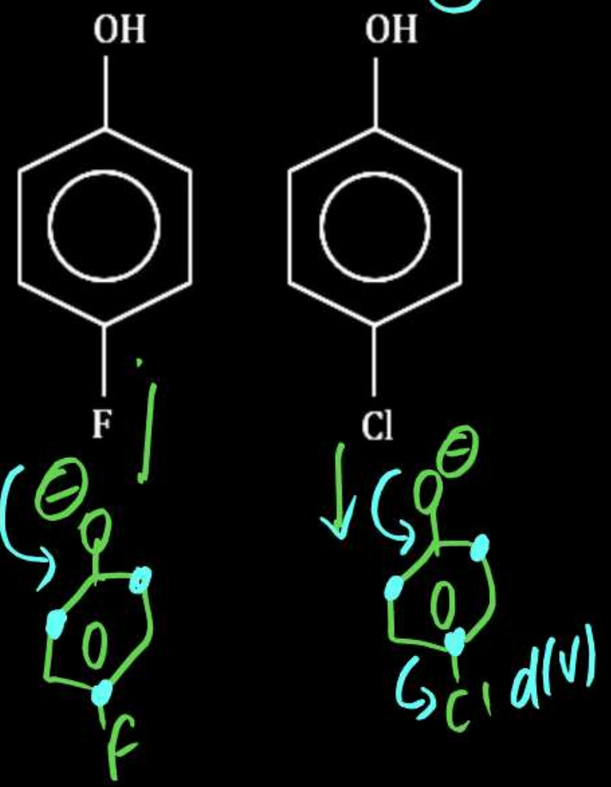
(43)



(III > II > I)

Imp

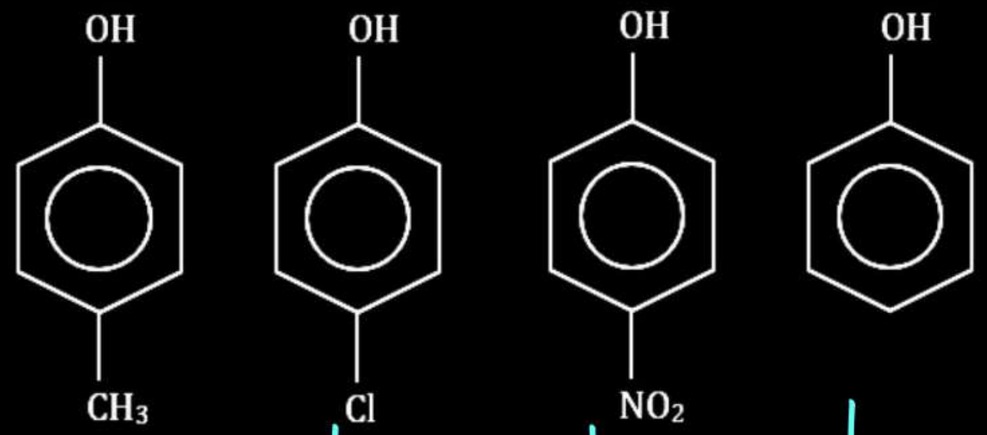
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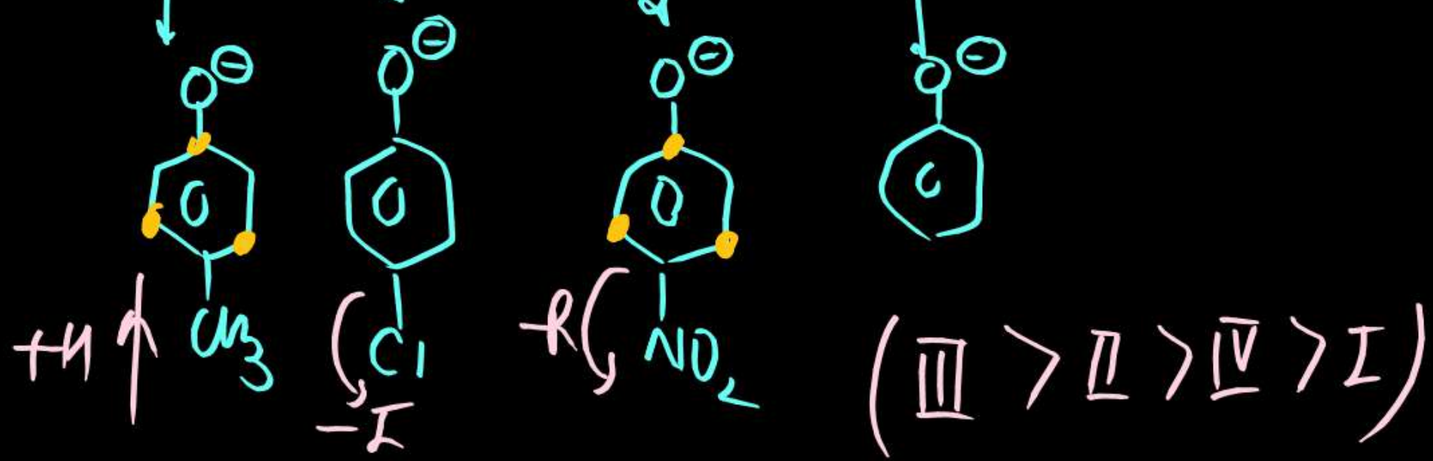


(II > I)

Organic Chemistry

(45)

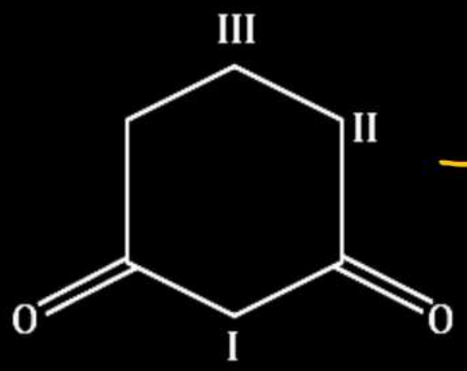




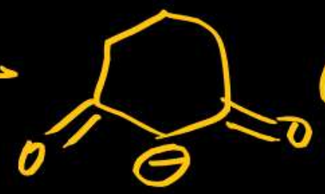
(III > II > IV > I)

Organic Chemistry

(46)

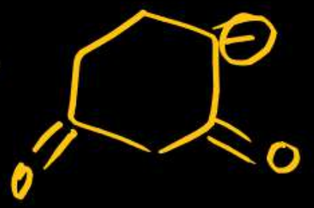


(I)



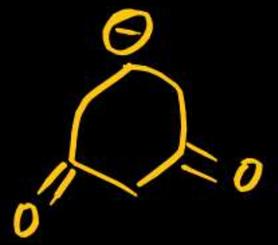
(3RS)

(II)



(2RS)

(III)

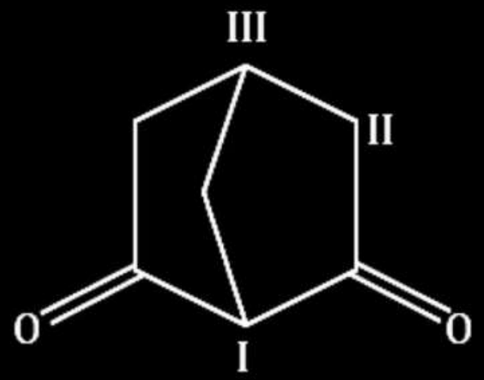


(0RS)

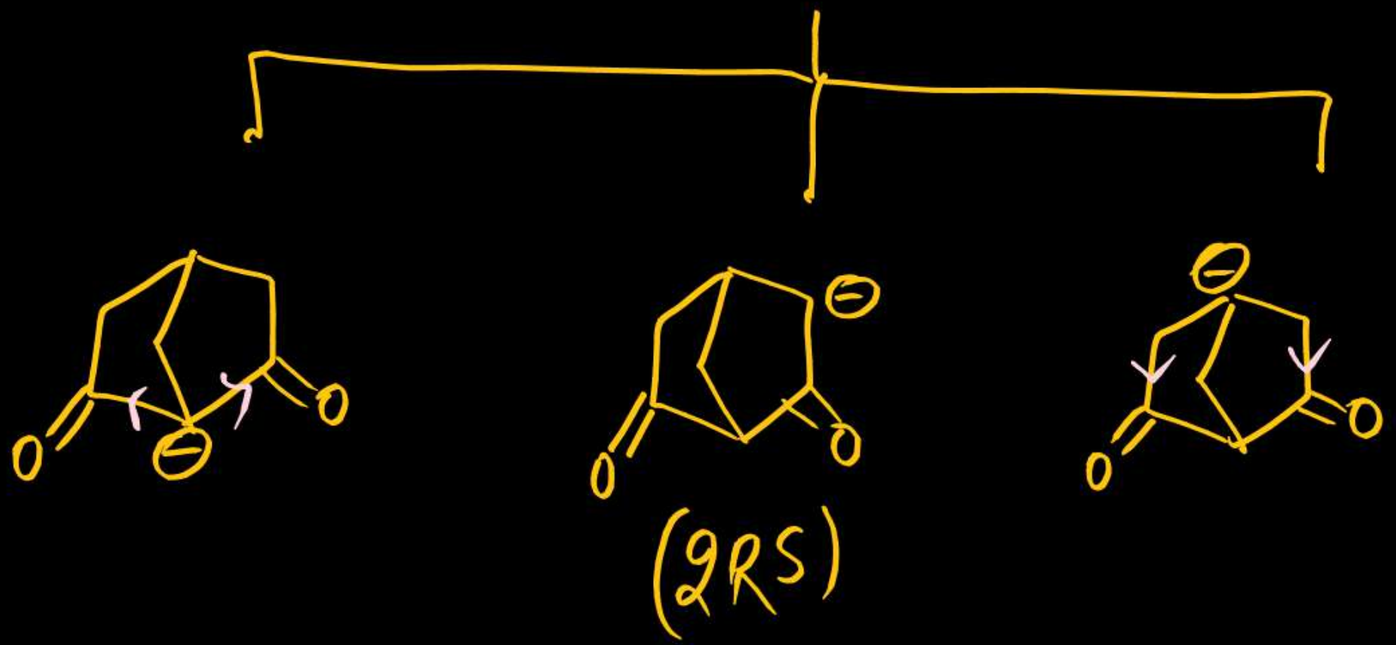
(I > II > III)

Organic Chemistry

(47)



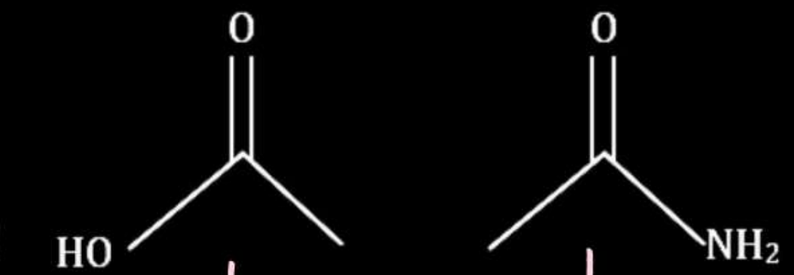
(II > I > III)



(2RS)

Organic Chemistry

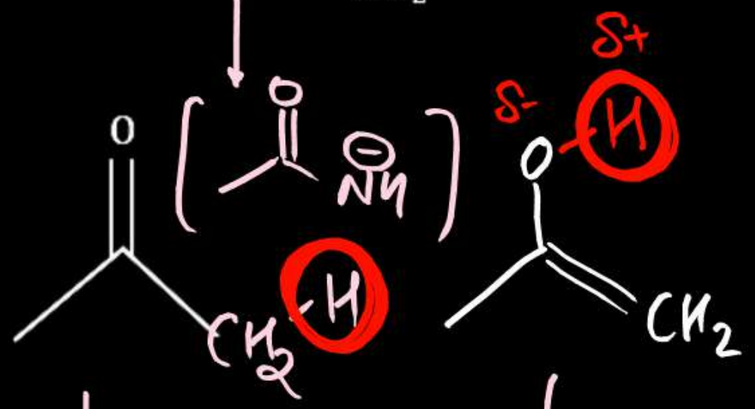
(48)



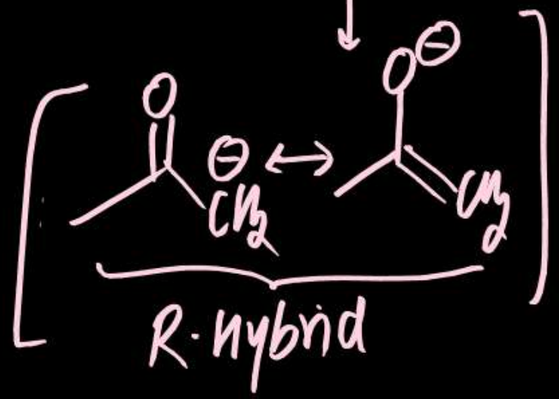
(I > II)

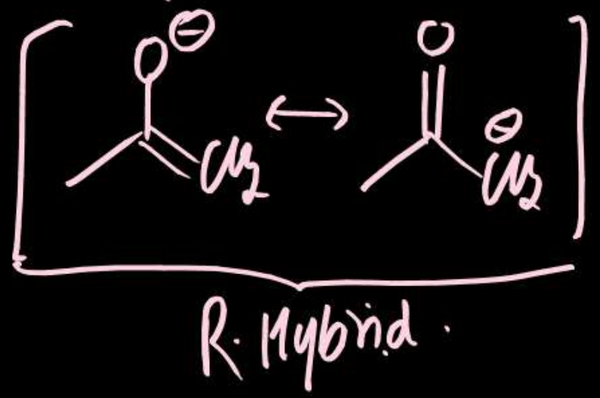
M.I.P

(49)



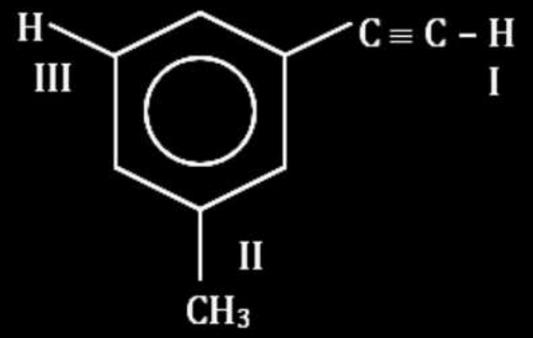
(II > I)



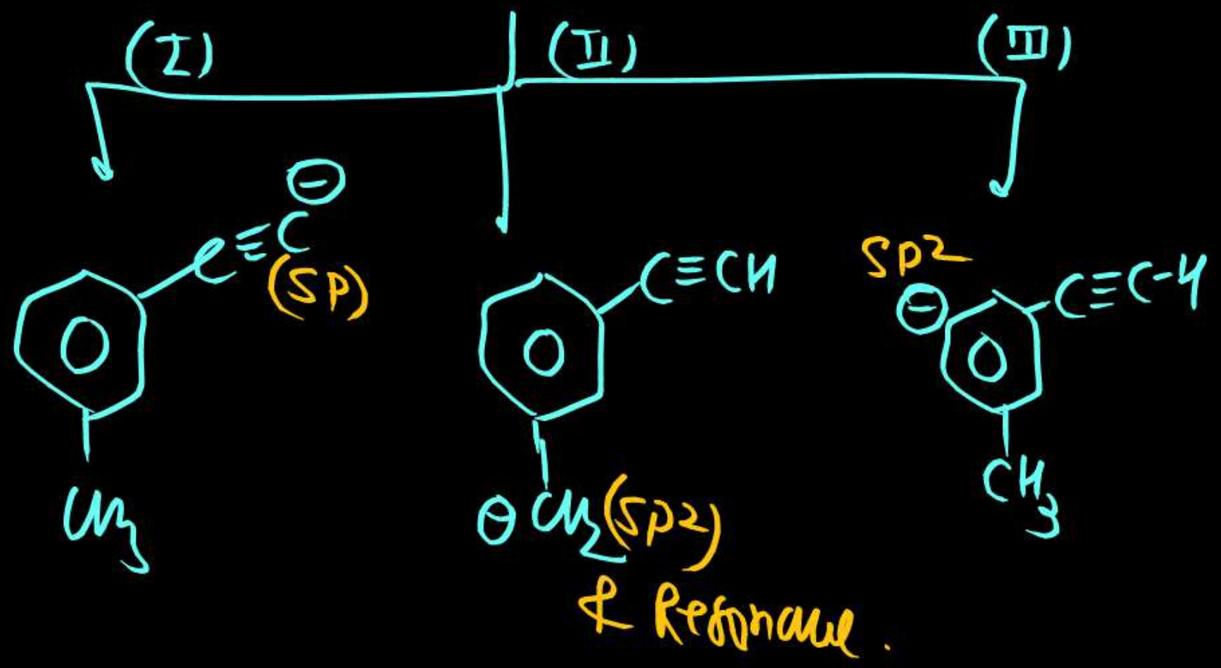


Organic Chemistry

(50)

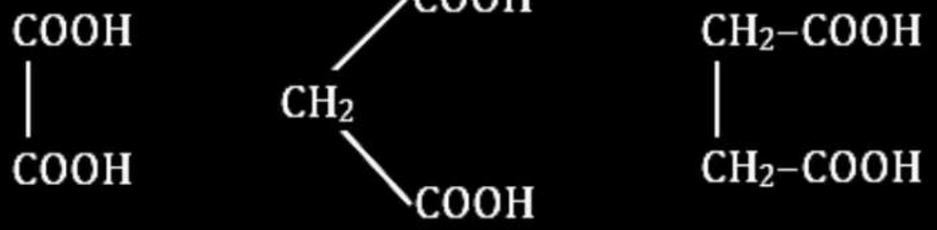


(I > II > III)



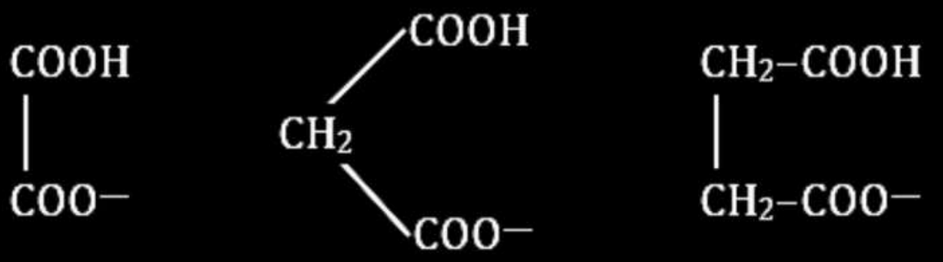
Organic Chemistry

(51)



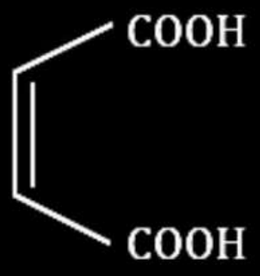
Organic Chemistry

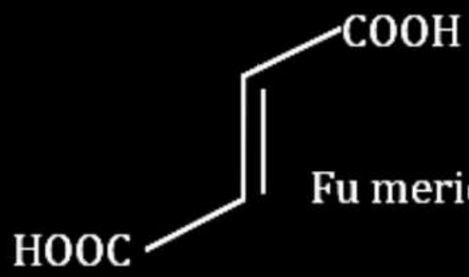
(52)



Organic Chemistry

(53)

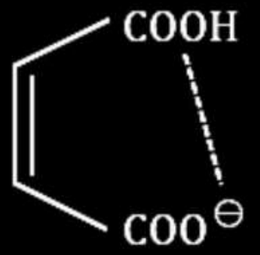


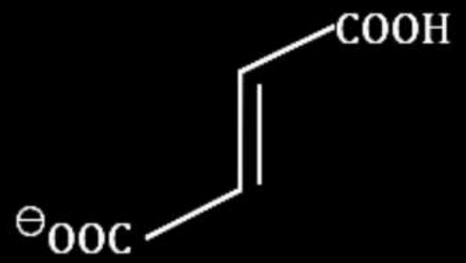


Fu meric Acid

Organic Chemistry

(54)

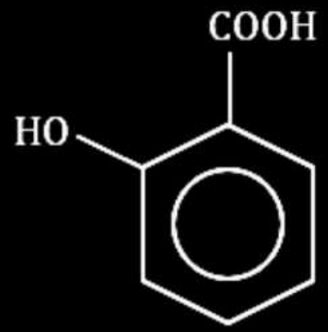


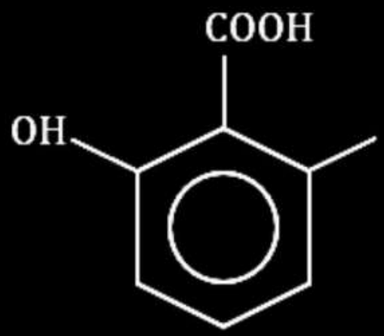


(55)



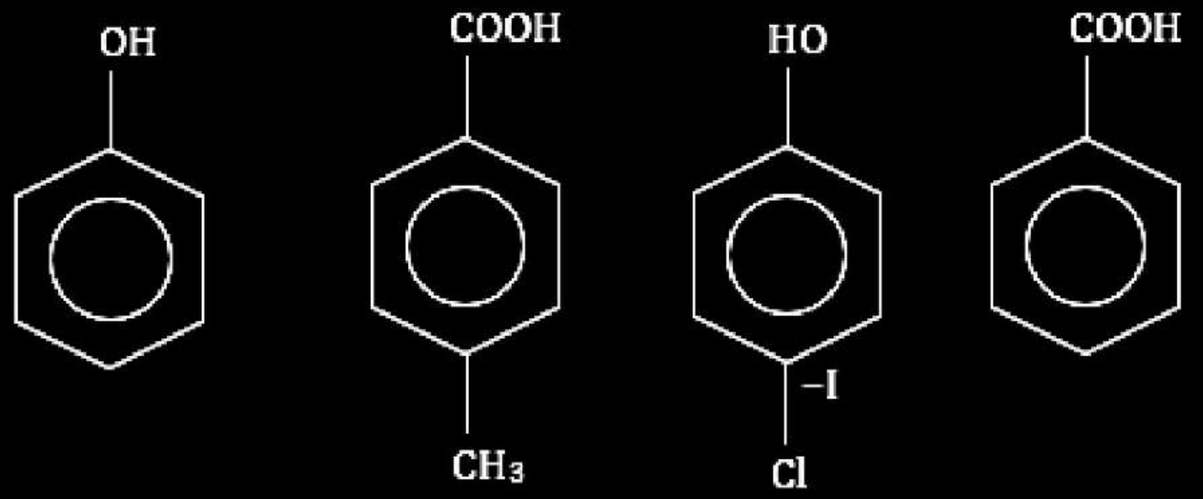






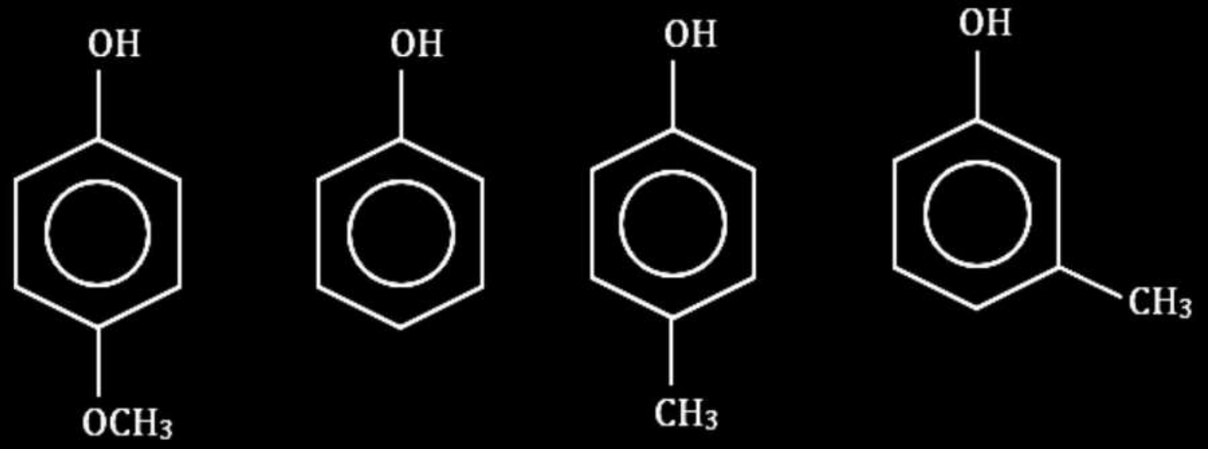
Organic Chemistry

(56)



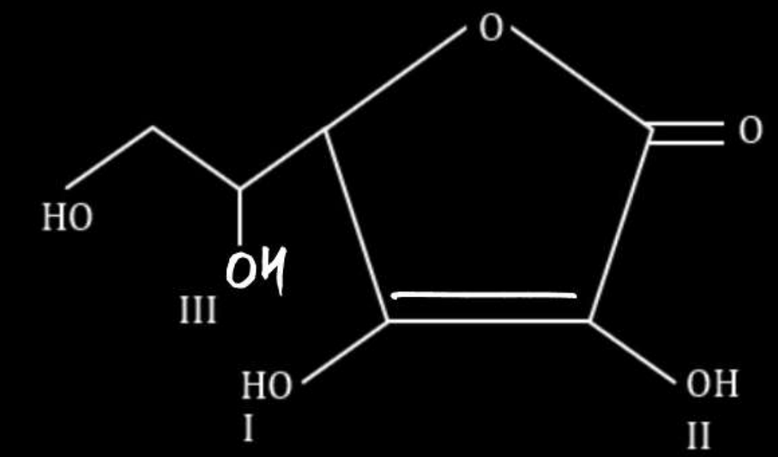
Organic Chemistry

(57)



Organic Chemistry

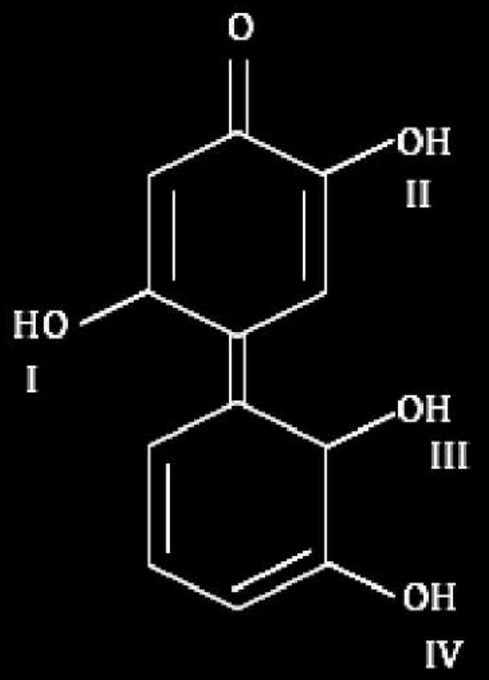
(58)



Ascorbic Acid (vitamin-C)

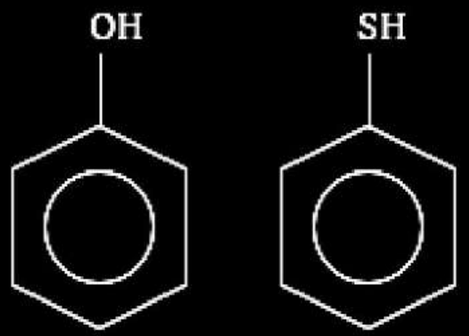
Organic Chemistry

(59)

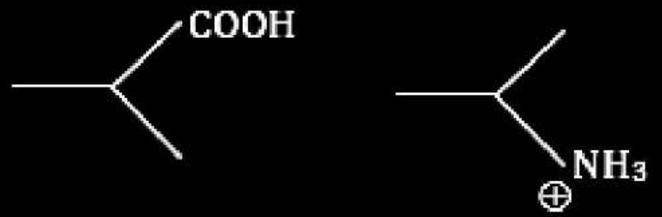


Organic Chemistry

(60)



(61)

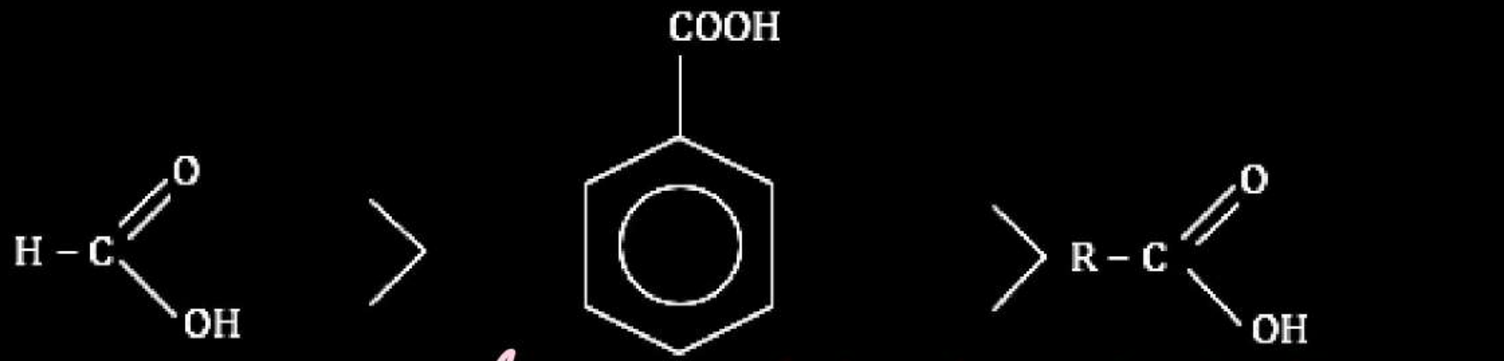


(62) $CH_3 - OH > H_2O > R - OH$

Organic Chemistry

MIup / M.Iup!-

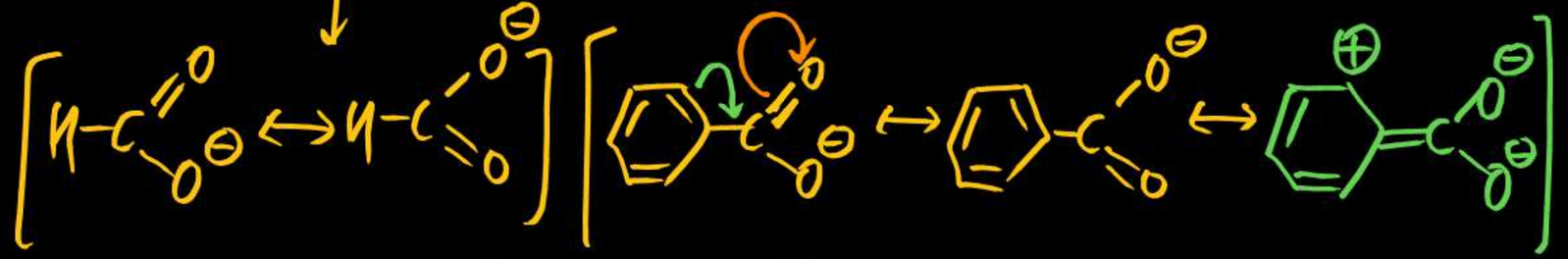
(63)



Aliphatic Carboxylic Acid

Aromatic Carboxylic Acid

Aliphatic Carboxylic Acid

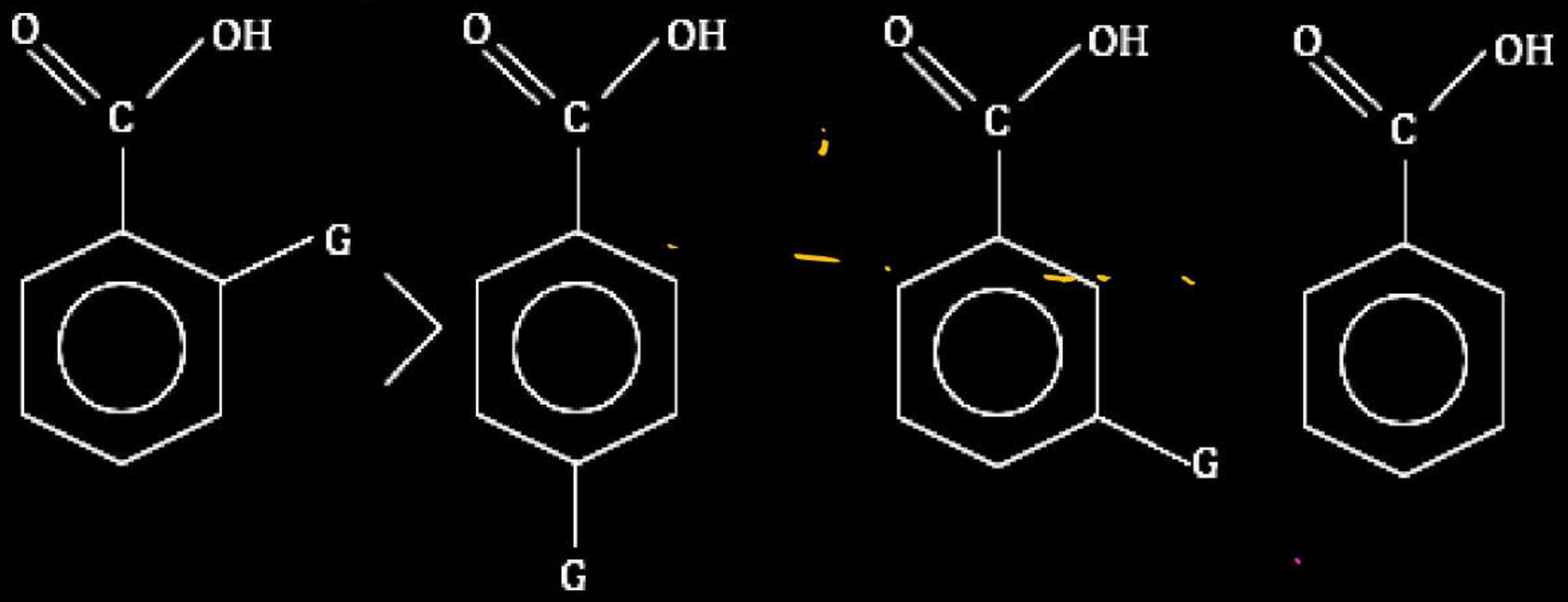


Note: H-COOH is only Aliphatic Acid which is more Acidic than Aromatic Carboxylic Acid.

Organic Chemistry

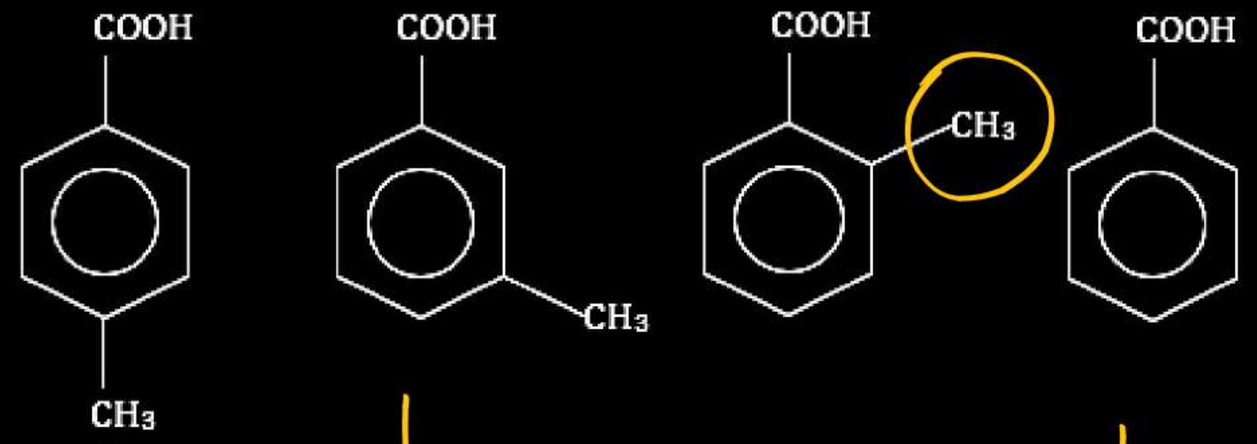
Ortho Effect :-

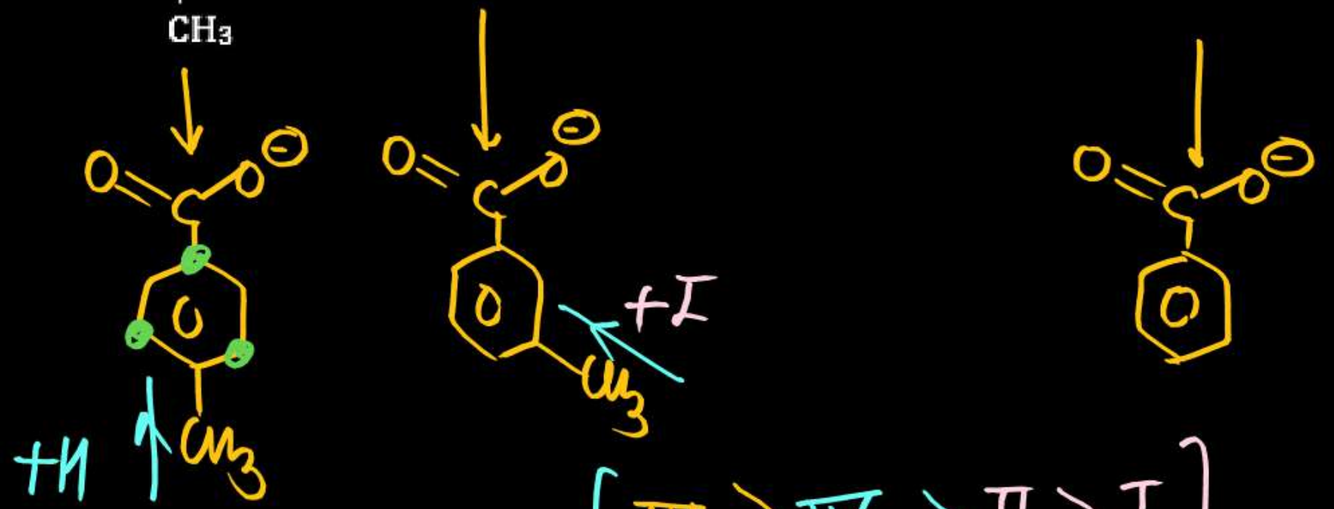
Ortho Substituted aromatic carboxylic Acid is more acidic than para, meta & non substituted aromatic Acid.



Organic Chemistry

(64)

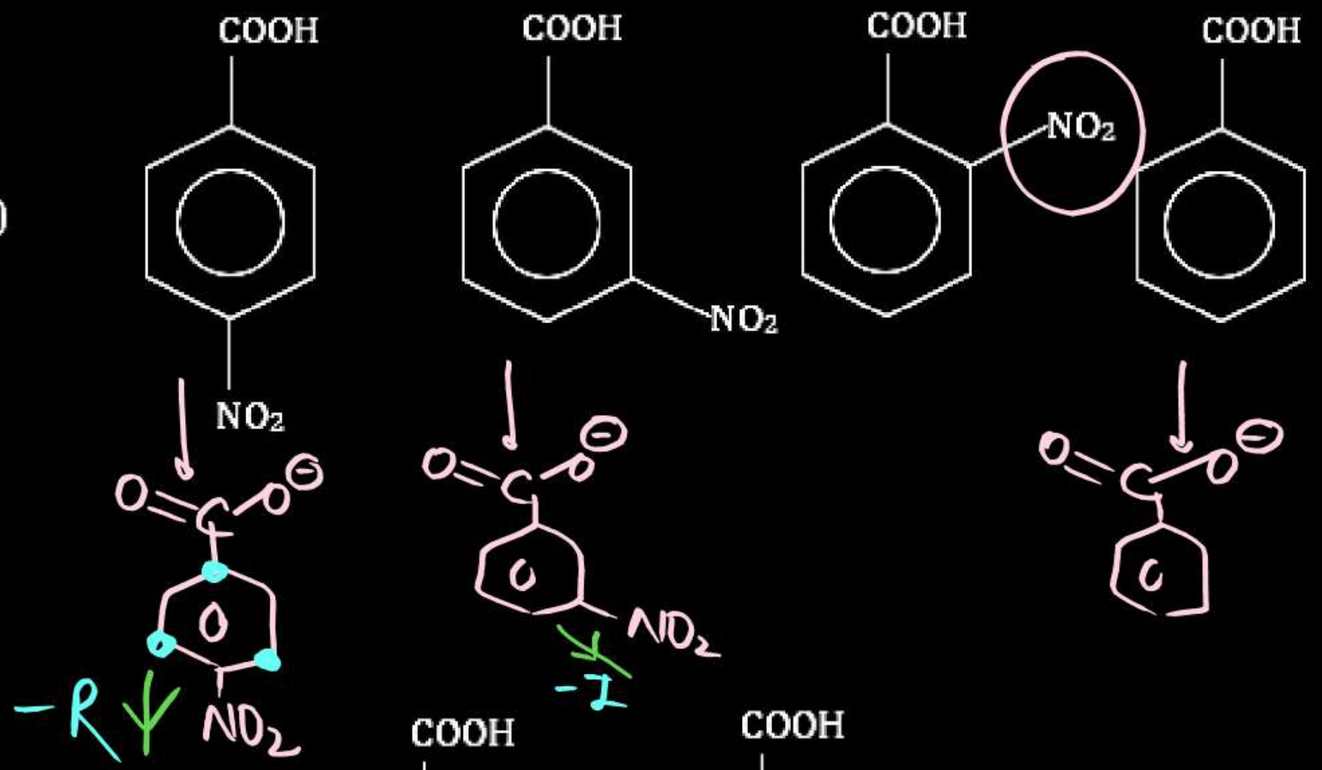




(III > IV > II > I)

Organic Chemistry

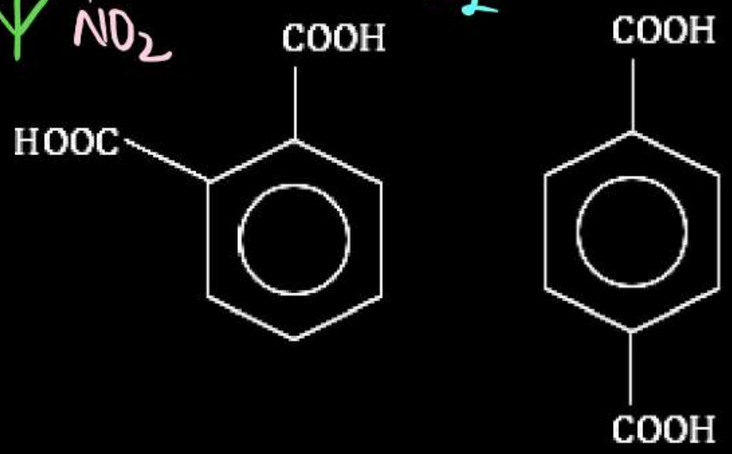
(65)



III > I > II > IV

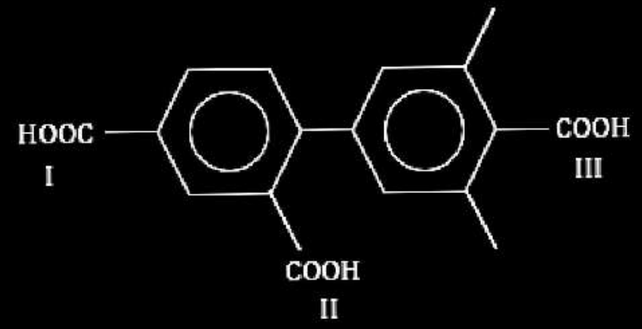
ortho effect -R -I

(66)

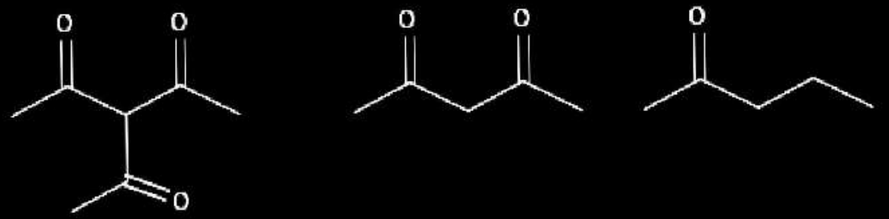


Organic Chemistry

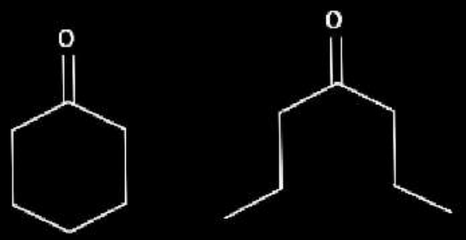
(67)



(68)

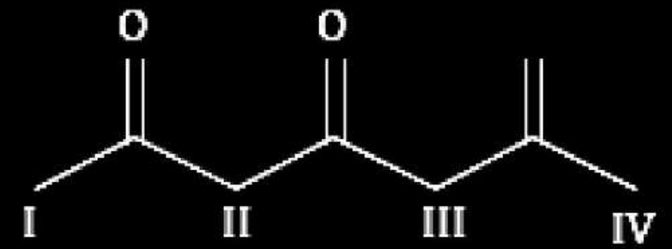


(69)

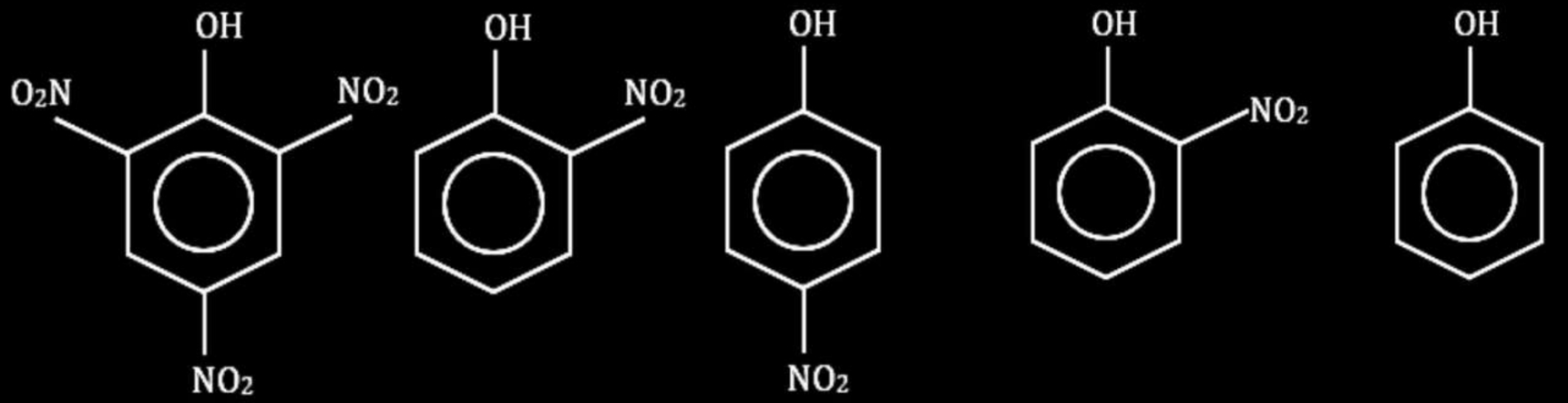


Organic Chemistry

(70)

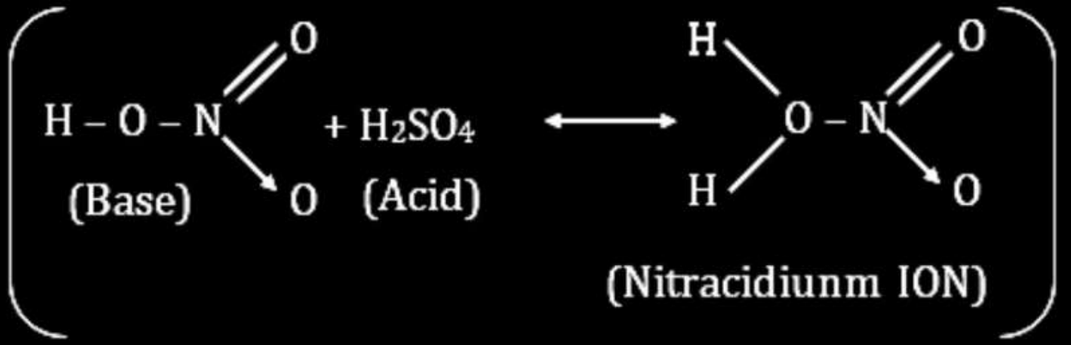


(71)



Organic Chemistry

(72) A mixture contains HNO_3 & H_2SO_4 write possible Reaction.



Organic Chemistry

(73) which of the following do not have Carboxy (–COOH) group.

(i) Vinegar (Acetic Arid)

(ii) carbolic Acid

(iii) Benzene Sulphonic Acid

(iv) Picric Acid

(v) Squaric Acid

Organic Chemistry

(vi) Oxalic Anid

(vii) Malonic Acid

(viii) Succinic Acid

(ix) Gluturic Acid

(x) Adipic Acid

Organic Chemistry

(xi) Palimitic Acid

(xii) Maleic Acid

(xiii) Fumeric Acid

(xiv) Malic Acid

(xv) Tartaric Aid

(xvi) lactic Acid

(xvii) Pyruvic Acid

(xviii) Citric Acid

(xix) Salicylic Acid

(xx) Cinamic Acid

(xxi) Barbiutaric Acid

(xxii) Ascorbic Acid

(xxiii) Aspartic Acid

(74)



less stable

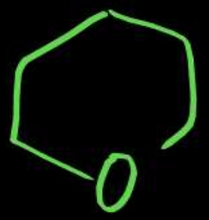


Stable

(I > II)

$-H^+$ $-H^+$

(Conjugate Base)

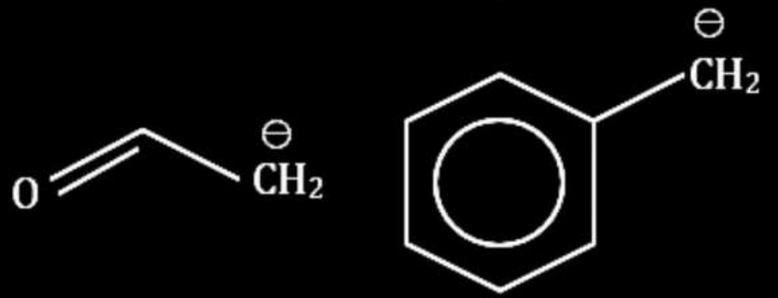




Organic Chemistry

(1) 

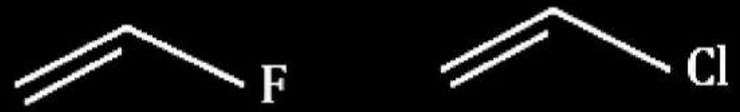
(2) 

(3) 

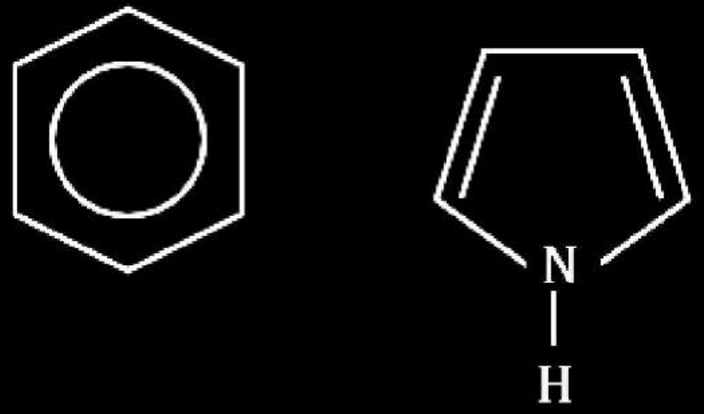
(4) 

Organic Chemistry

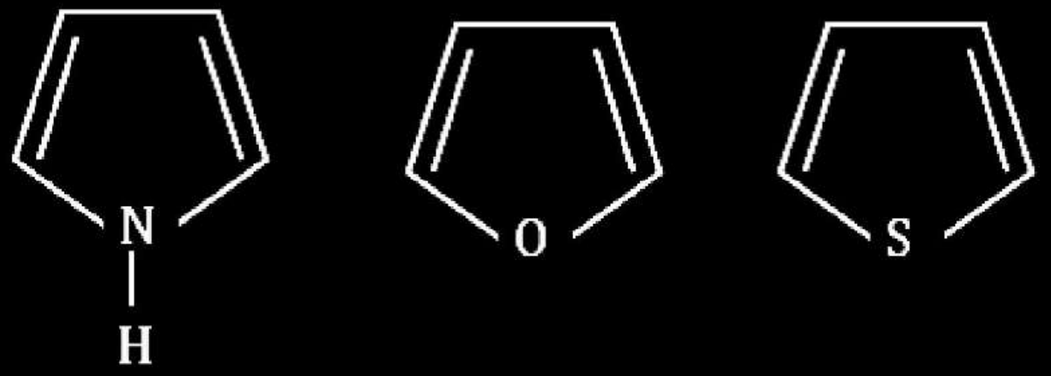
(5)



(6)

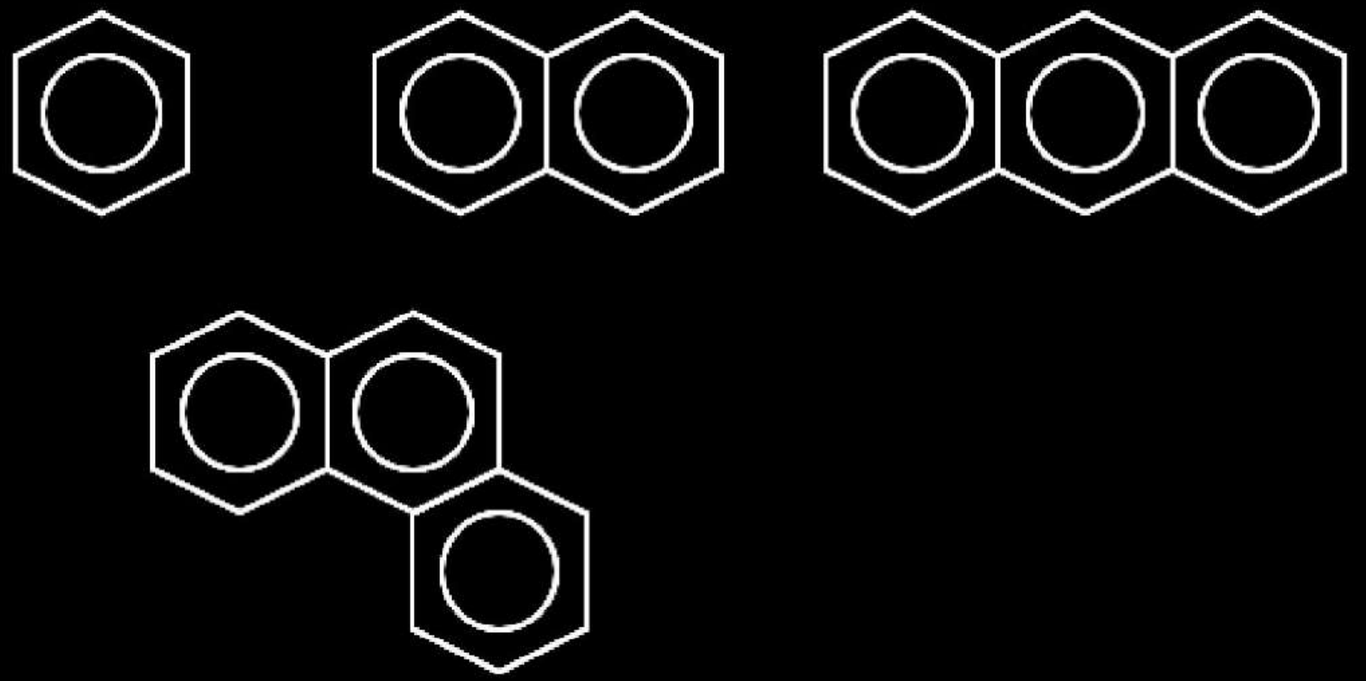


(7)



Organic Chemistry

(8)



Basic strength :-

Arrange following in decreasing order of Basic strength

less stable/most Basic — most stable

(1) (Base) $F^{\ominus}$ $Cl^{\ominus}$ $Br^{\ominus}$ $I^{\ominus}$ less Basic

H^+ H^+ H^+ H^+

Conjugate Acid :- H—F H—Cl H—Br H—I (Neutral)

Most stable — Least stable

Basic strength $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

Organic Chemistry

(2) $R\text{–}O^{\ominus}$ $R\text{–}S^{\ominus}$

(3)

$CH_3^{\ominus}$	$NH_2^{\ominus}$	$OH^{\ominus}$	$F^{\ominus}$
↓	↓	↓	↓
C. Acid CH_4	NH_3	H_2O	HF

Organic Chemistry

M.Imp:

To Compare Basic strength

(*) Aliphatic Amine > Aromatic Amine

(*) Localised e^- density > delocalized e^- density

(*) Conjugate Acid stability

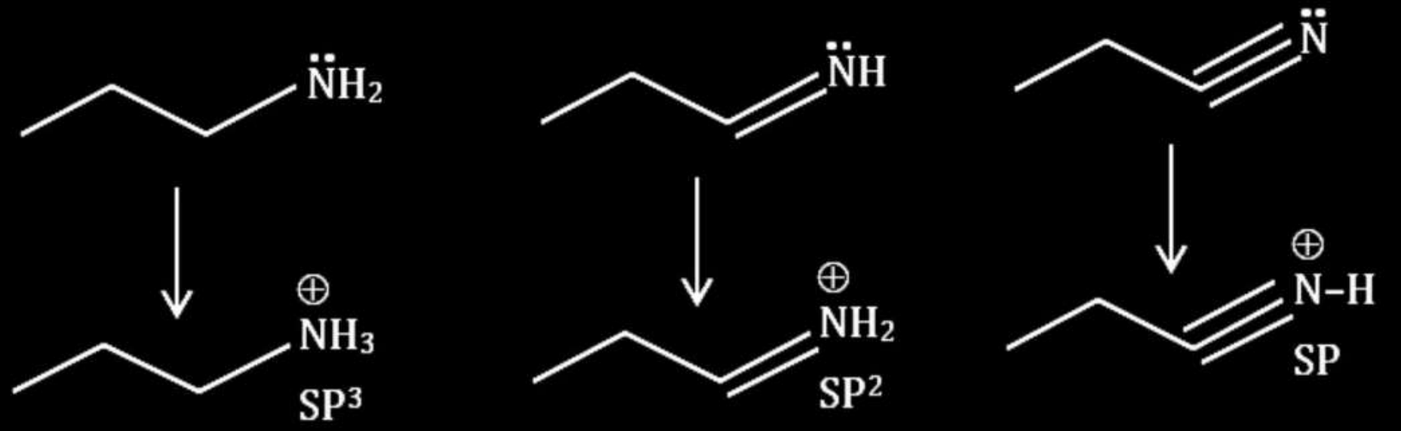
- By resonance
- Hybridisation ($Sp^3 > Sp^2 > Sp$)
- Solvation effect (Ortho Subs. Amine weakest Base)

(*) EDG > EWG

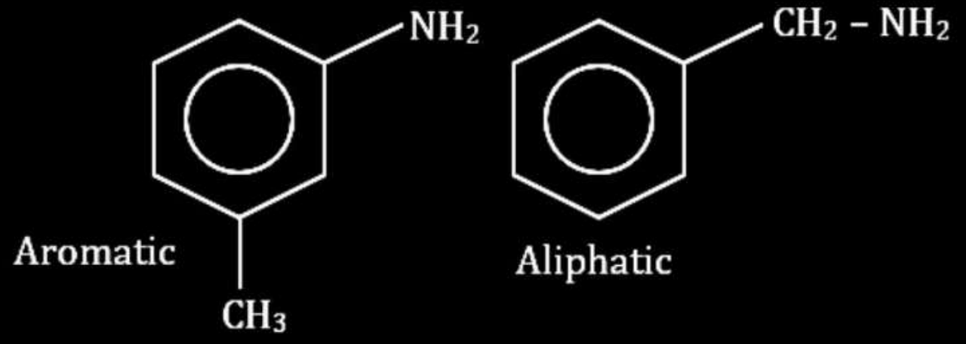
Organic Chemistry

(4) $CH_3 - \overset{\ominus}{C}H_2$ $\qquad$ $CH_2 - \overset{\ominus}{C}H$ $\qquad$ $HC \equiv \overset{\ominus}{C}$

(5)

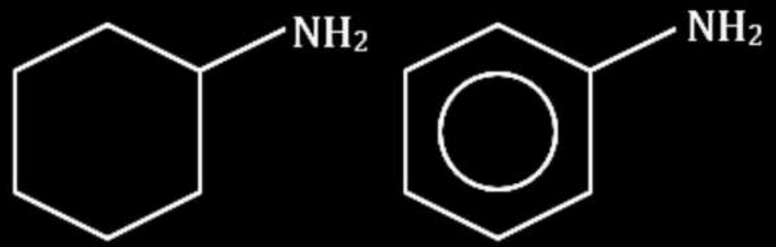


(6)

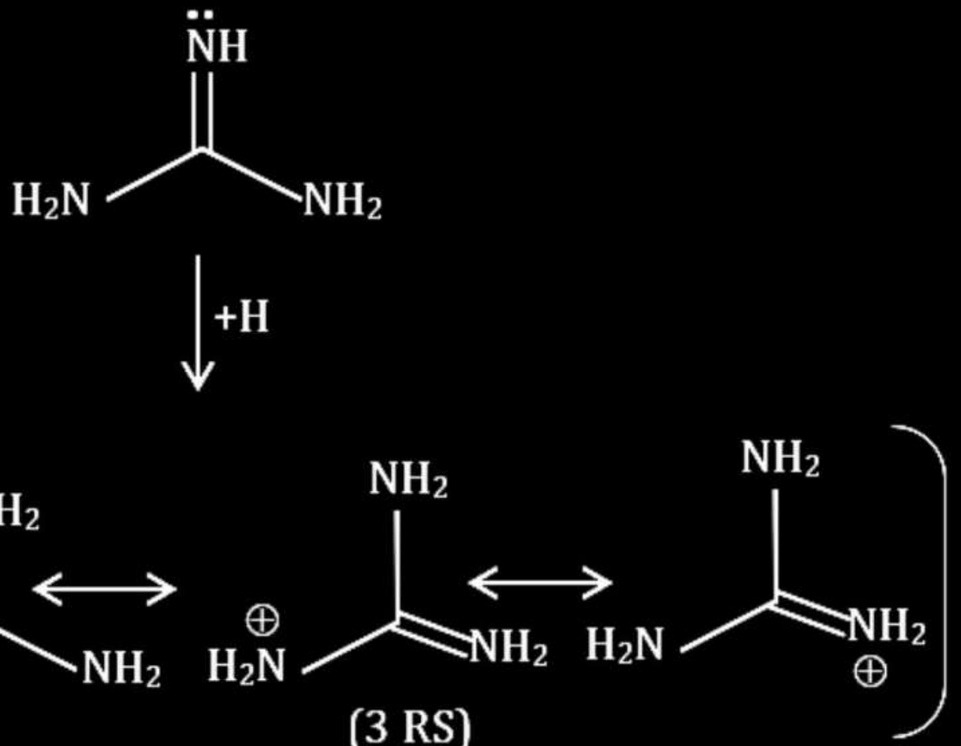


Organic Chemistry

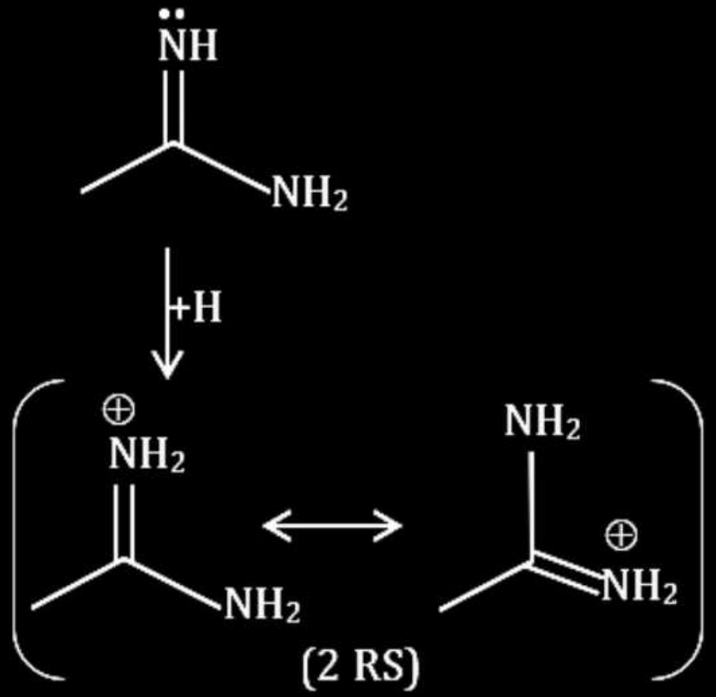
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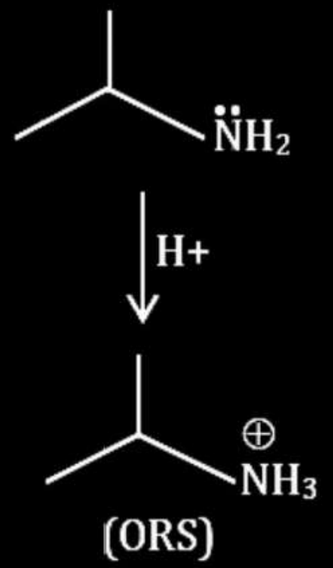


(8)

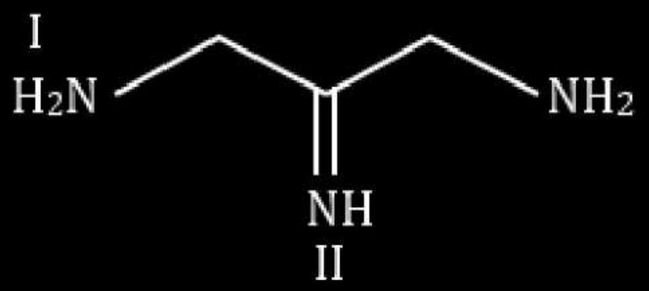


(3 RS)



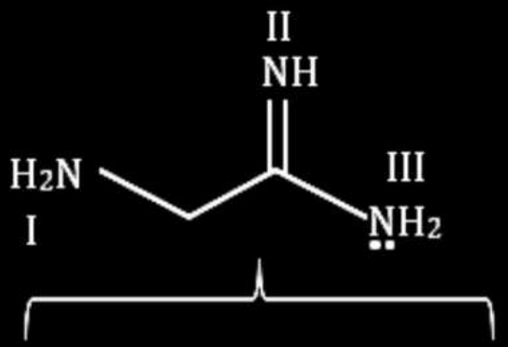


(9)

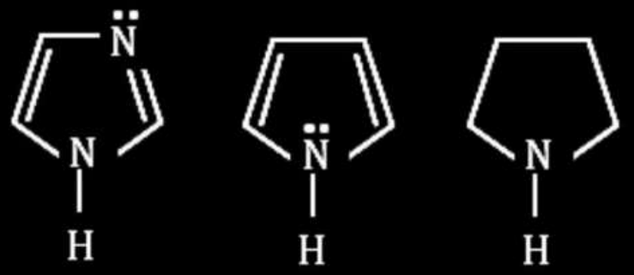


Organic Chemistry

(10)



(11)



Aromatic Amine

Aromatic Amine

Aliphatic Amine

(12)



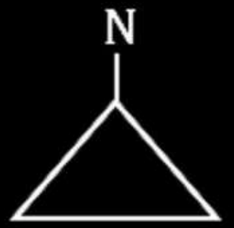
Organic Chemistry

(13)





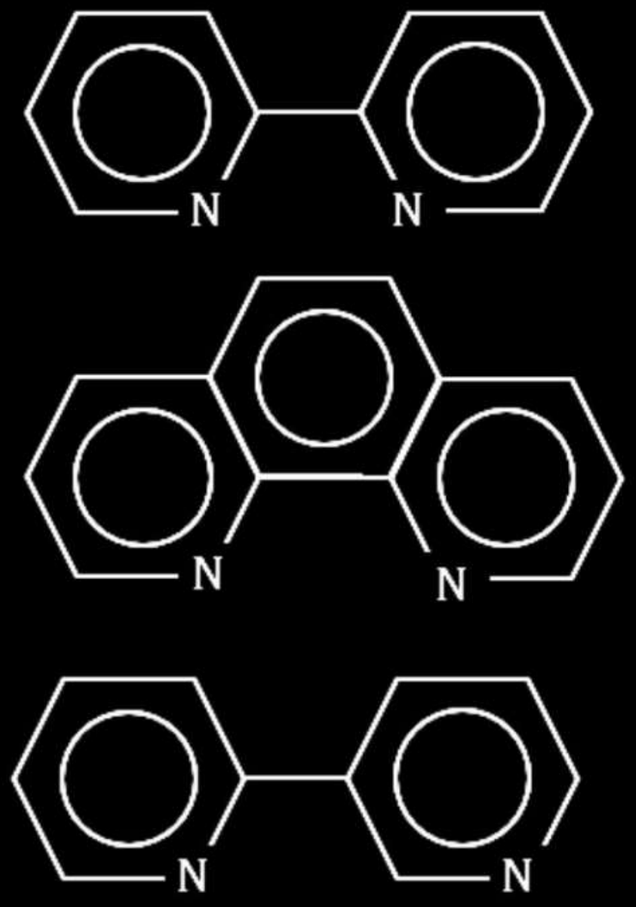
(14)





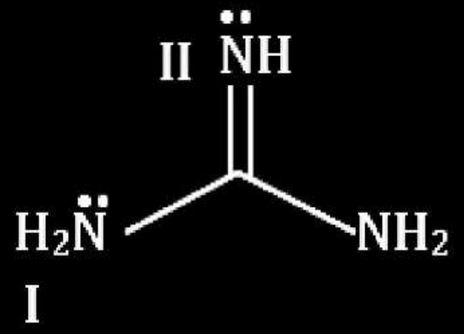
Organic Chemistry

(15)



Organic Chemistry

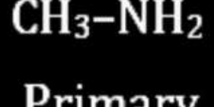
(16)



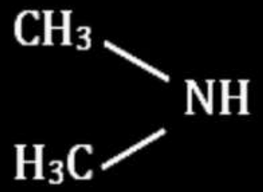
(17)



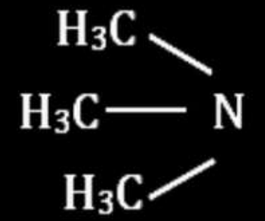
Ammonia



Primary Amine



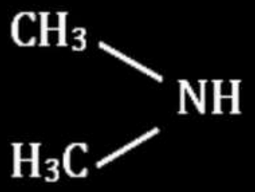
Sec. Amine

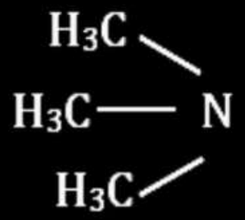


Terl. Amine

(in Aq. phase)
H_2O

(18)

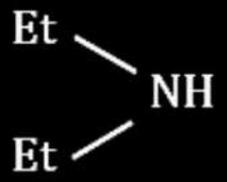


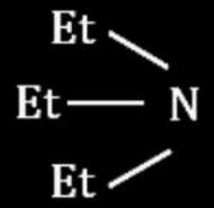


(in Gas phase)

Organic Chemistry

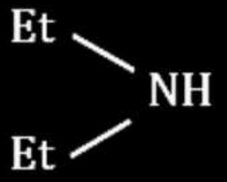
(19) H_3N $Et–NH_2$

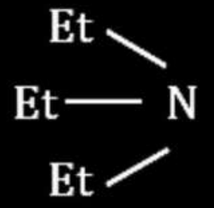




(in Gas phase)

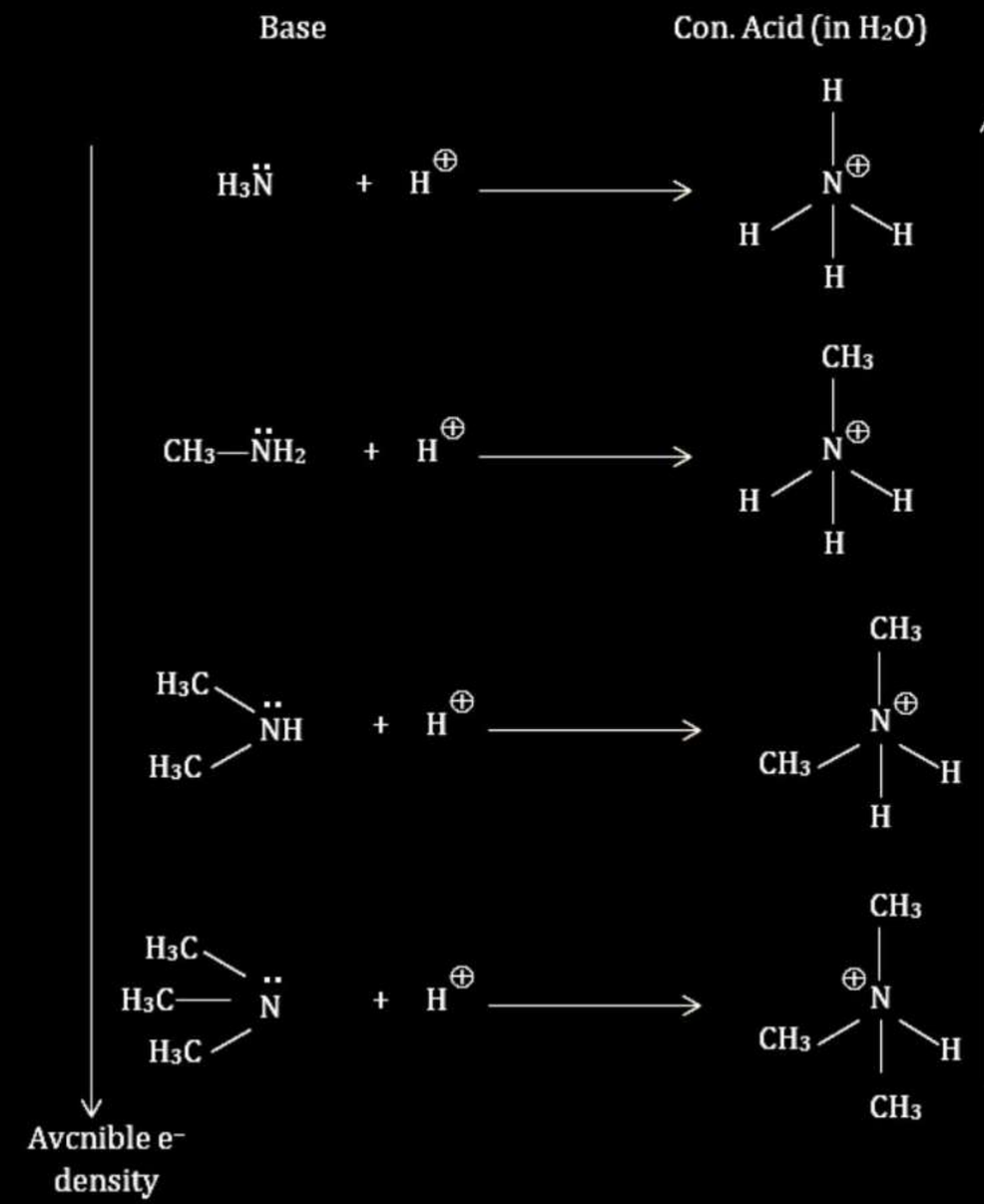
(20) H_3N $Et–NH_2$





(in H_2O)

Organic Chemistry



Base	Av e⁻ density	Ex. of solve
H_3N	⊠	↑↑↑
$Me–NH_2$	↑	↑↑
Me_2NH	↑ ↑	↑
Me3N	↑↑↑	⊠

$Me_2NH > Me–NH_2 > Ml_3N > NH_3$

Organic Chemistry

(21)

